

Renovascular Hypertension, Atherosclerotic Renal Vascular Disease, and Thromboembolic Kidney Disease

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NORMAL RENOVASCULAR ANATOMY

The clinical presentations of renovascular disease are influenced by the acuity, nature, and site of renal vascular compromise. In most individuals, the kidney has a single renal artery (RA) with a lumen diameter of 3 to 7 mm. The main RA branches in a double or triple fork pattern or ladder pattern.¹ The incidence of multiple RAs is about 31%, with bilateral supernumerary arteries in 11%.² The collaterals to the kidney, depicted in [Fig. 43.1](#), can maintain renal parenchymal viability in the face of main RA occlusion.

Renal veins (RVs) begin in the subcapsular region of the kidney, communicate with perirenal and cortical venous channels, and empty into interlobular and then arcuate veins. The venae rectae drain the pyramids and join the arcuate veins, which leave the renal parenchyma through interlobar vessels, converging into four to six trunks near the hilum of the kidney. Approximately 20% of people have two or more RVs. The main RVs empty into the inferior vena cava. The left RV is three times longer than the right (7.5 cm vs. 2.5 cm) and traverses behind the splenic vein and body of the pancreas before it crosses in front of the aorta near its termination at the inferior vena cava. About 25% of people have retroaortic or circumaortic RVs. RV anatomy is shown in [Fig. 43.2](#).

CLINICAL SYNDROMES ASSOCIATED WITH RENAL VASCULAR DISEASE

Impairment of blood flow is associated with one or more distinct clinical syndromes. These are summarized in [Box 43.1](#). Discussion of these syndromes as distinct entities is meant to help with recognition and understanding that often, in clinical practice, there is significant overlap. The first section will focus on the three most common clinical presentations of RA stenosis: renovascular hypertension (RVH), atherosclerotic renovascular disease (ARVD; formerly called ischemic renal disease [IRD]), and unstable cardiac syndromes including flash pulmonary edema. The second section will address the distinct clinical presentations of transplant renovascular disease, renal infarction, athroembolic disease, and RV thrombosis.

RENOVASCULAR HYPERTENSION

A reduction in renal perfusion pressure activates a series of hormonal and neuronal responses that raise systemic arterial pressure to restore

RA perfusion pressures.³ Renovascular hypertension (RVH) is defined as a syndrome of elevated blood pressure (BP) produced by any condition that leads to reduced perfusion of the kidneys.

Central to the pathogenesis of RVH is activation of the renin-angiotensin-aldosterone system (RAAS) with release of renin from the juxtaglomerular apparatus. This is mediated in part by stimulation of neuronal nitric oxide synthase and cyclooxygenase-2 in the macula densa. Blockade of RAAS during creation of an experimental RA stenosis prevents development of hypertension.⁴ Studies in transgenic mice without receptors for angiotensin II (Ang II) confirm that development of RVH requires an intact RAAS.⁵ In the absence of RAAS blockade, systemic arterial pressures increase until renal perfusion is restored. Studies in experimental models and humans indicate that additional mechanisms contribute to long-term BP elevation in the presence of RA stenosis, including activation of the sympathetic nervous system, impairment of nitric oxide generation, release of endothelin, and hypertensive microvascular injury to the kidney.⁶

Mechanisms responsible for sustained RVH differ depending on whether one or both kidneys are affected; these mechanisms have been studied in animal models in which RA perfusion is reduced by clipping the vessel proximally. The nomenclature distinguishes between a situation in which one clip is present with a normal contralateral unclipped kidney ("1-clip-2-kidney hypertension") and a situation in which the entire renal mass is affected ("1-clip-1-kidney hypertension"). Both these situations begin with impaired renal perfusion and initial activation of RAAS with sodium retention. However, in the 1-clip-2-kidney hypertension model, the elevated pressure generated by RAAS activation mediates a pressure natriuresis in the nonstenotic kidney. This restores plasma volume and results in sustained hypoperfusion and RAAS activation in the poststenotic kidney. This sequence of events produces Ang II-dependent hypertension ([Fig. 43.3A](#)).

By contrast, 1-clip-1-kidney hypertension represents a model in which the entire kidney mass is exposed to reduced perfusion pressure. As a result, sodium retention leads to expanded blood volume and sustained elevation in pressure, which then restores renal perfusion pressure beyond the stenosis and inhibits RAAS activation (see [Fig. 43.3B](#)). Hypertension in this model is typically more volume dependent than Ang II dependent. It may be less responsive to RAAS blockade than 1-clip-2-kidney hypertension. [Box 43.2](#) lists causes of RVH based on these mechanisms.

Renal Collateral Blood Supply

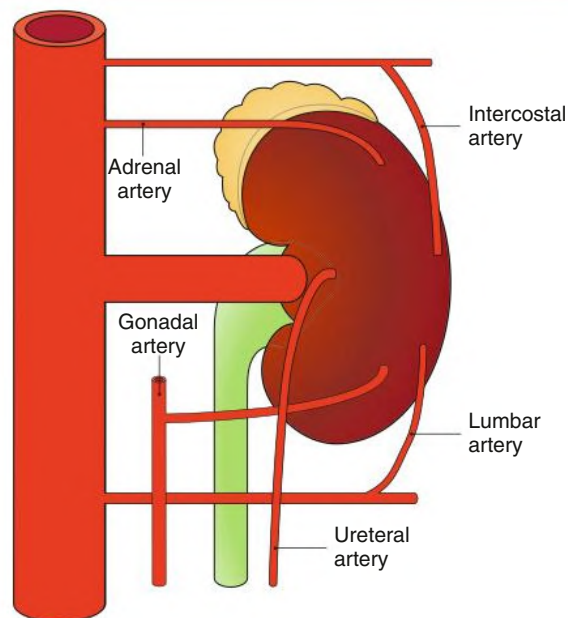


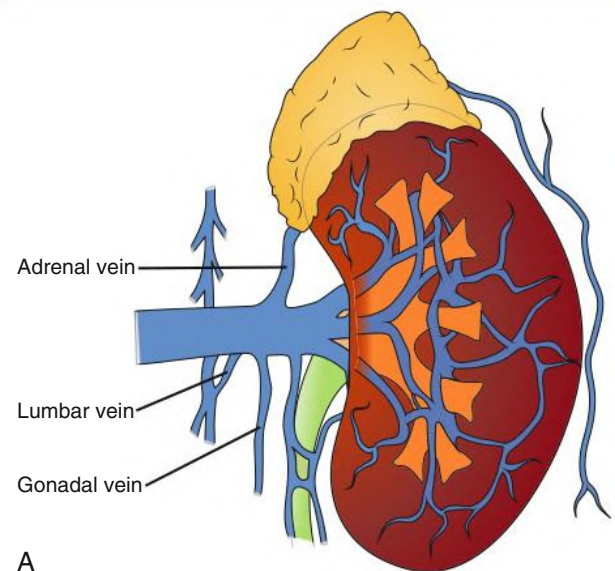
Fig. 43.1 The potential collateral renal arterial vessels to the kidney.

In RA stenosis, RAAS activation occurs when luminal stenosis is relatively severe, usually in the range of at least 60% to 80% cross-sectional diameter narrowing (Fig. 43.4). In experimental models, the relative importance of pressor mechanisms, including measurable activation of the RAAS, changes with time. Levels of circulating plasma renin activity tend to decrease. Several mechanisms have been proposed to explain such changes, including a slowly developing pressor action of Ang II, transition to alternative pressor mechanisms, and hypertensive injury to the nonstenotic kidney. In experimental models, this translates into a time limit for reversibility of RVH by removal of the clip. Clinically, this makes it difficult to determine when patients are most likely to benefit from RA revascularization for cure of hypertension.

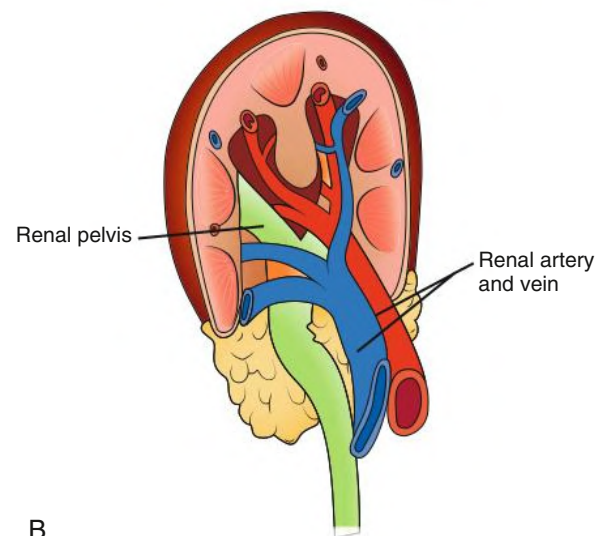
Widespread use of RAAS blockers to manage hypertension, congestive heart failure (CHF), and diabetic and other proteinuric nephropathies has increased the exposure of individuals with undetected RVH to these agents. Therefore, many cases of RVH go undiagnosed unless hypertension becomes more difficult to control or kidney dysfunction develops. Patients who typically undergo diagnostic evaluation for RA stenosis are a subset of patients with more severe or resistant hypertension or those presenting with renal impairment or cardiac syndromes.

Clinical differentiation of RVH and primary hypertension is difficult, and they may be superimposed. If secondary to unilateral RA stenosis, for example, RVH often can be easily controlled with the use of RAAS blockers. Certainly, some cases of RVH present with accelerated, resistant, or hypertension emergency. Clinical studies suggest that for any level of BP, patients with RVH have higher nocturnal pressures (“nondippers”) and have more severe target organ manifestations, such as left ventricular hypertrophy (LVH) and albuminuria, than patients with primary hypertension.⁷ Patients with RVH may present with hypokalemia and metabolic alkalosis, which are clues to secondary aldosteronism. Clinical suspicion for RVH arises when hypertension develops either very early (<30 years) or later in life (>70 years) or when well-controlled hypertension becomes more resistant. Renovascular hypertension may rarely be associated with renin-mediated and hemodynamically induced nephrotic range proteinuria

Renal Vein Anatomy



A



B

Fig. 43.2 Renal Vein Anatomy. (A) There is extensive communication between the renal venous plexus and lumbar, gonadal, and adrenal veins, which provide alternative outflow in the setting of renal vein thrombosis, particularly on the left. (B) Transverse section of the kidney showing relative position of vascular structures in the renal pelvis. (From Graham SD, Keane TE, Glenn JF, eds. *Glenn's Urologic Surgery*. 7th ed. Wolters Kluwer/Lippincott Williams & Wilkins Health; 2010.)

BOX 43.1 Clinical Presentations of Renovascular Disease

- Renovascular hypertension
- Ischemic renal disease
- Unstable cardiac syndromes
- Renal infarction
- Atheroembolic renal disease
- Renal vein thrombosis
- Transplant renovascular disease

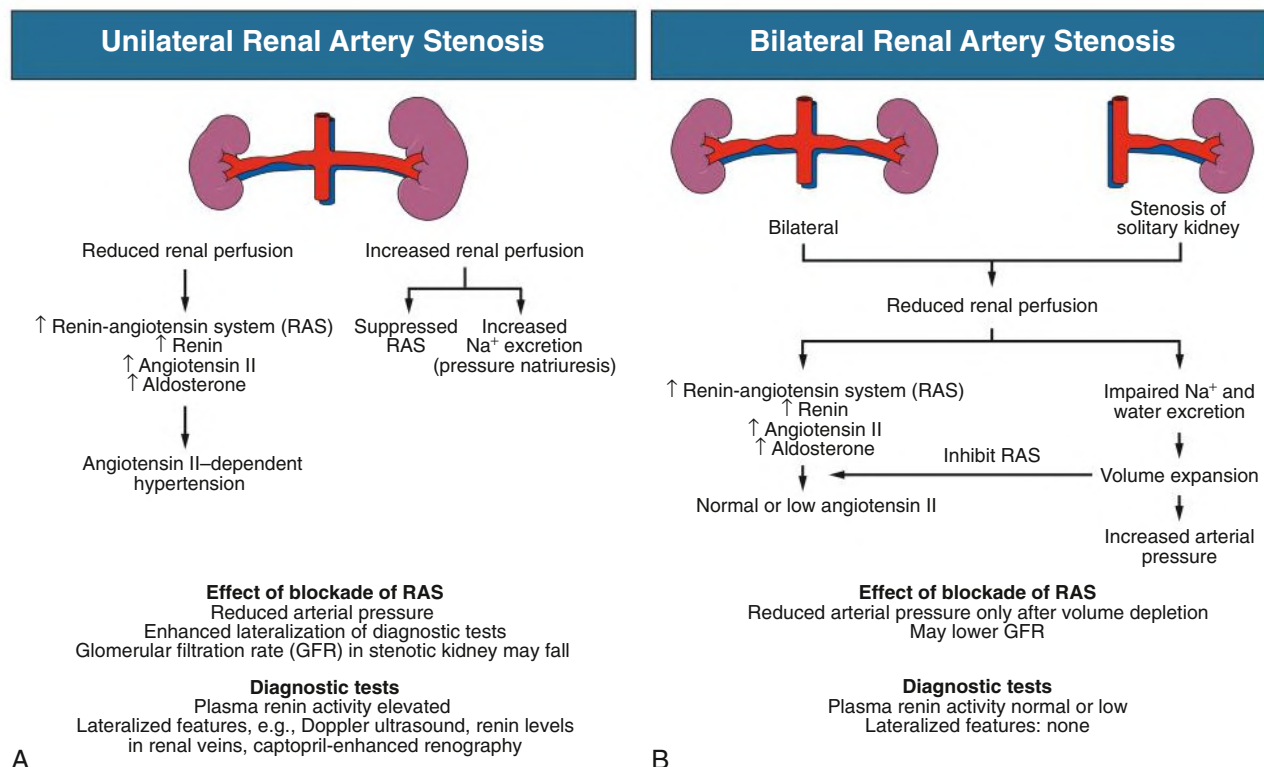


Fig. 43.3 Pathogenesis of Renovascular Hypertension in One-Kidney Versus Two-Kidney Model. (A) In unilateral stenosis with two kidneys, opposing forces between the stenotic kidney, which has reduced perfusion pressures, and the nonstenotic contralateral kidney, which has increased perfusion pressures, result in laboratory and clinical features of angiotensin-dependent hypertension. (B) In unilateral stenosis with a solitary functioning kidney or in a patient with bilateral critical renal artery stenosis, reduced perfusion pressure to the stenotic kidney in the absence of a normal kidney excreting sodium leads to sodium and volume retention, ultimately associated with hypertension without persistent activation of the renin-angiotensin system (RAS). *GFR*, Glomerular filtration rate.

that resolves with treatment. A syndrome of polydipsia associated with hyponatremia attributed to the dipsogenic action of Ang II has also been observed in patients with RVH. Clinical features of RVH are summarized in [Box 43.3](#).

Renal Artery Stenosis

The most common cause of RVH is RA stenosis. Fibromuscular dysplasia (FMD) and atherosclerotic disease are the two most common causes of RVH. The unique clinical presentations of congenital and inflammatory arteritis as exemplified by aortic coarctation and Takayasu arteritis will also be discussed.

Fibromuscular Dysplasia

Fibromuscular dysplasia is the most common cause of RVH in children and young adults and is defined as an idiopathic, segmental, non-inflammatory, nonatherosclerotic disease of the muscular arterial wall. Fibromuscular dysplasia is defined by pathognomonic arteriographic abnormalities, most commonly either a focal stenosis involving any portion of the vessel or multifocal stenoses in a string of beads appearance ([Fig. 43.5](#)) in mid to distal portions of small and medium-sized arteries. Nonstenotic angiographic findings associated with FMD include aneurysms, dissection, or vessel tortuosity, the latter often seen in the carotid circulation as an S-curve or in the coronary circulation. The presence of focal or multifocal stenosis in at least one vascular bed is necessary for the diagnosis of FMD, with the recognition that tortuosity, aneurysm, or dissection in another vascular bed may be associated but not sufficient for the diagnosis.⁸

The vascular distribution of FMD involves primarily the renal and extracranial cerebral and vertebral arteries but may involve any artery. In the FMD registries, RAs are involved with FMD in 66% to 92% of cases.⁸ Bilateral RA involvement is seen in 25% to 35% of adult cases. Cerebrovascular involvement was identified in 80% and 59% of patients in the US and European registries, respectively.^{8,9} Evaluation of 489 patients for RA and cerebral vascular involvement found that 48% had two territories involved, and 66.1% had dysplastic stenosis, aneurysms, or dissections at another site.¹⁰ Of those with imaging of other vascular beds reported in the US registry, 48% had coronary involvement and 45% had lower extremity vascular involvement. Less common extrarenal sites of involvement with FMD include mesenteric, celiac, splenic, and aortic arteries.

Given the potentially systemic nature of FMD, we recommend thorough imaging assessment with computed tomographic angiography (CTA) or magnetic resonance angiography (MRA) of the entire vasculature at least once in patients with FMD to look for involvement at other sites, including occult aneurysms.⁸

The prevalence of renovascular FMD is estimated at 4 in 1000.¹¹ Angiography data from potential kidney donors suggest that the prevalence may be higher, with FMD observed in 3.8% to 6.6% of individuals.¹² Approximately 80% to 90% of cases of FMD occur in women, with most of these having multifocal FMD. Men with FMD tend to have a more aggressive phenotype, with a greater prevalence of focal FMD and aneurysms.¹³

Of those enrolled in the US FMD registry, 91% are White, and the mean age of onset of hypertension is 44.8 years. The mean age at

diagnosis was 53.3 years. As this presentation overlaps with the age of onset of primary hypertension, FMD may be missed in many adults. This racial predilection could represent recruitment or geographic bias.

BOX 43.2 Causes of Renovascular Hypertension

Two-Kidney Hypertension^a

- Unilateral fibromuscular dysplasia
- Unilateral atherosclerotic renovascular disease
- RA aneurysm
- RA embolism and infarction
- Traumatic arterial occlusion
- Arteriovenous fistula
- RA dissection or thrombosis
- Aortic dissections with compromise to renal ostium
- Page kidney
- Takayasu arteritis
- Metastatic tumor compressing renal parenchyma
- Pheochromocytoma compressing RA
- Phakomatosis pigmentovascularis type IIb
- Neurofibromatosis
- Behçet disease
- Covering of origin of RA by aortic stent graft
- RA spasm

One-Kidney Hypertension^b

- Stenosis to solitary kidney
- Bilateral arterial stenosis or dissection
- Coarctation of the aorta
- Vasculitis involving RAs
- Congenital vascular anomalies
- Atheroembolic renal disease

^aTwo-kidney hypertension implies that a contralateral, nonaffected kidney is present.

^bOne-kidney hypertension implies that the entire renal mass is beyond the vascular lesion, either bilateral disease or a solitary functioning kidney.
RA, Renal artery.

Histologically, FMD is associated with distortion of the architecture of the arterial wall involving one or more layers. The former distinction between adventitial, medial, intimal, and perimedial variants is no longer considered clinically relevant. Both genetic and environmental factors have been implicated in the etiology of FMD. In current registries, less than 7.5% of cases have family history of FMD. Data are biased by the significant percentage of asymptomatic cases and underdiagnosis. Genome-wide association studies have identified a single nucleotide polymorphism in the PHACTR1 locus on chromosome 6 as a risk allele. This genetic location is associated with regulation of expression of endothelin 1, which has effects on vascular tone and remodeling.¹⁴ The current consensus is that there is no genetic test specific for FMD, and relatives of patients with FMD should be screened clinically and with imaging if signs, including hypertension, or symptoms develop. Environmental factors implicated in the etiology of FMD include cigarette smoking, hormonal influences, and vascular trauma or stretching of the RA during development or associated with nephropthosis.¹³ US registry data observed a higher incidence of aneurysms and major vascular events among smokers with FMD.¹⁵ Ongoing research is exploring potential roles of circulating transforming growth factor- β (TGF- β) and proinflammatory lipid mediators in the vascular wall in the pathogenesis of FMD.¹⁶

BOX 43.3 Clinical Features of Renovascular Hypertension

- Activation of renin-angiotensin system (early)
- Early-onset (<30 years) or late-onset (>60 years) hypertension
- Activation of sympathetic nervous system
- Abnormal circadian rhythm: loss of nocturnal fall
- Secondary aldosteronism: hypokalemia
- Accelerated target organ damage
- Microvascular disease
- Left ventricular hypertrophy
- Reduced glomerular filtration rate
- Hyponatremic hypertensive syndrome
- Unstable cardiac syndromes
- Rarely, nephrotic range proteinuria

Hemodynamic Effects of Stenotic Lesions in Renal Arteries

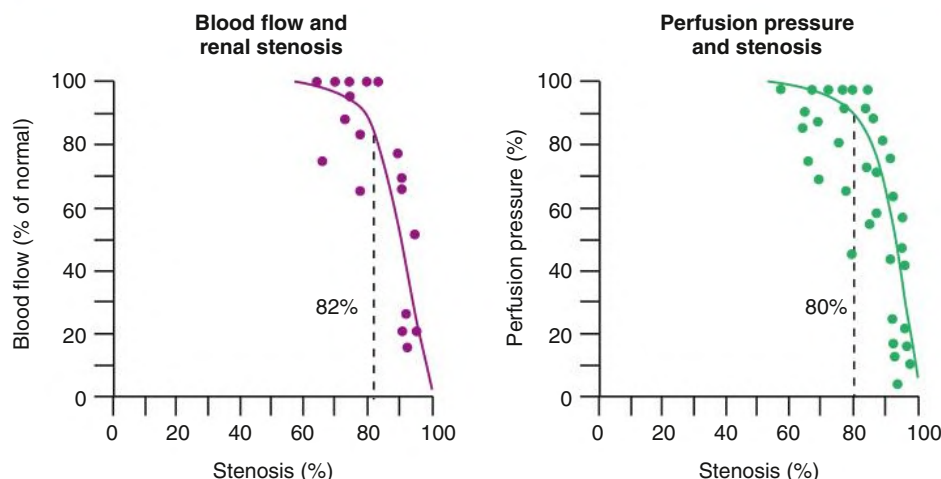


Fig. 43.4 Hemodynamic Effects of Stenotic Lesions in Renal Arteries. Changes in blood flow and arterial pressure across a carefully quantitated arterial lesion are barely detectable until cross-sectional area diminishes by 75% to 80%.

The most common adult clinical presentation is early-onset RVH in young to middle-aged women. Cerebrovascular lesions may manifest with headaches, pulsatile tinnitus, and bruits over the carotid arteries. Fibromuscular dysplasia should be considered in younger patients presenting with hypertension and stroke, transient ischemic attacks, subarachnoid hemorrhage, or amaurosis fugax. Clinical presentations of FMD and associated disorders are shown in [Box 43.4](#).

The natural history of FMD has not been adequately studied. Progression of disease may manifest as new lesions within the same arterial bed, worsening arterial luminal narrowing within a specific lesion, involvement of a new vascular territory, or development or enlargement of arteriovenous fistulas or aneurysms. Up to 37% of patients may demonstrate angiographic progression of FMD, with few patients developing new or progressive lesions after the age of 40 years.¹⁷ Progression may be associated with kidney cortical thinning but rarely causes advanced kidney failure.

Atherosclerotic Renal Artery Stenosis

Atherosclerotic RA stenosis is the most common cause of RVH in patients over 50 years old. Estimates of the prevalence of atherosclerotic RA stenosis depend on the population screened. One population-based study of 870 patients over 65 years old and screened with RA duplex ultrasound found a 6.8% prevalence of atherosclerotic RA stenosis, defined as greater than 60% stenosis.¹⁸ Autopsy series report an overall prevalence of 4% to 20%, with progressively higher prevalence for those older than 60 years (25%–30%) and older than 75 years (40%–60%). Atherosclerotic RA stenosis occurs in patients with more generalized atherosclerosis involving the aorta, peripheral vasculature, and coronary arteries. Approximately 14% of patients with hypertension and 20% of patients with hypertension and diabetes mellitus (DM) have evidence of atherosclerotic RA stenosis. Coexistent RA stenosis is found in 11% to 30% of patients undergoing coronary angiography

and in up to 45% of patients undergoing peripheral angiograms.¹⁹ Among patients with chronic CHF, RA stenosis has been reported in 50% to 68%.²⁰ The American College of Cardiology and American Heart Association recommend screening for atherosclerotic RA stenosis in patients presenting with resistant or accelerated hypertension and those presenting with heart failure.

Predictors of RA stenosis include a history of hypertension, presence of chronic kidney disease (CKD), coexisting peripheral vascular or coronary artery disease, the presence of abdominal bruits, and a history of smoking. Atherosclerotic RA stenosis is bilateral in 20% to 40% of patients. In many of these cases, the degree of stenosis is below the threshold to cause activation of RAAS or have other clinical implications. However, the presence of atherosclerotic RA stenosis is a marker for increased cardiovascular and mortality risk.²¹ Accordingly, its presence should prompt clinical attention to cardiovascular risk factors, including the use of high-dose statin therapy, efforts to promote smoking cessation, and optimal control of BP and metabolic syndrome.

Some patients with atherosclerotic RA stenosis will experience progressive RA luminal narrowing and develop RVH or other clinical syndromes described later. Prospective studies between 1990 and 1997 using duplex ultrasound in patients with atherosclerotic RA stenosis indicated progression in 30% over 3 years, varying by degree of initial stenosis, with progression more common in those with more than 60% luminal diameter stenosis. Total occlusion is rare but devastating, often leading to loss of kidney function. There are few prospective data assessing progression in patients treated with optimal medical therapy targeting atherosclerotic risk factors in the modern era.

Clinical presentations and treatment of atherosclerotic RA stenosis are discussed later in this chapter.

Takayasu Arteritis

Initially described in 1761, Takayasu arteritis, often termed *pulseless disease*, is one of several inflammatory disorders involving the renal

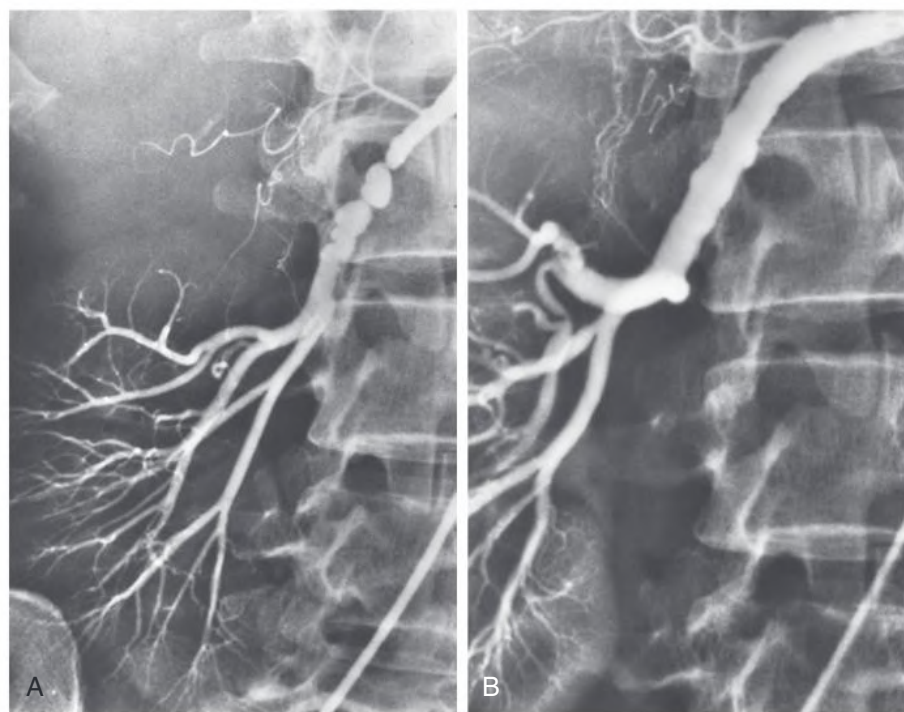


Fig. 43.5 Fibromuscular Dysplasia. (A) Selective renal arteriogram illustrating the beaded appearance of fibromuscular dysplasia with multiple webs characteristic of multifocal fibromuscular dysplasia in a 39-year-old woman. (B) Selective injection of the same renal artery after technically successful percutaneous transluminal renal angioplasty. (Courtesy Michael McKusick, MD, Mayo Clinic, Rochester, MN.)

BOX 43.4 Clinical Manifestations and Disorders Associated With Fibromuscular Dysplasia

Clinical Manifestations

- Incidental finding (e.g., living kidney donors)
- Renovascular hypertension
- Renal infarction
- Loin or flank pain
- Hematuria
- Retroperitoneal hemorrhage
- Cerebrovascular accident (stroke)
- Transient ischemic attack
- Headaches
- Pulsatile tinnitus
- Horner syndrome
- Neck pain
- Dizziness
- Imaging finding of aneurysm dissection
- Amaurosis fugax
- Myocardial infarction
- Ischemic chest pain/dyspnea
- Postprandial abdominal pain
- Weight loss
- Hemobilia
- Claudication

Associated Disorders

- Tuberous sclerosis
- Marfan syndrome
- Ehlers-Danlos syndrome
- Cystic medial necrosis
- Coarctation of the aorta
- Alport syndrome
- Renal agenesis or dysgenesis
- α_1 -Antitrypsin deficiency
- Medullary sponge kidney
- Cigarette smoking
- Collagen III glomerulopathy
- Atherosclerotic renovascular disease
- Alagille syndrome
- Ask-Upmark kidney
- Celiac disease
- Cocaine exposure (intrauterine)
- Crohn disease
- Homocystinuria
- Macrophagic myofasciitis
- Neurofibromatosis
- Williams syndrome
- Pheochromocytoma
- Infantile myofibromatosis
- Ergotamine preparation, methysergide

vasculature causing RVH. Although rare in the United States, its prevalence varies geographically, with reports as high as 1 in 3000 in Japan.²² Takayasu arteritis usually presents between the ages of 25 and 41 years. The diagnosis should be considered in any child or young adult with hypertension and/or asymmetric peripheral pulses. In half of cases, identification of arterial stenosis is preceded by a prodromal illness characterized by fever, night sweats, malaise, and weight loss. Inflammatory markers are often elevated during this phase. Later,

vascular stenoses can lead to RVH, kidney dysfunction, stroke, cerebral hemorrhage, myocardial infarction, or CHF, depending on sites of involvement.²²

The diagnosis of Takayasu arteritis requires arteriographic narrowing or occlusion of a vascular territory of the aorta, its branches, or large arteries not attributable to atherosclerosis, middle aortic syndrome, or FMD.²³ One distinguishing feature of Takayasu arteritis is the presence of inflammatory thickening of the vascular wall seen on imaging.²³

Theories about the pathophysiology of Takayasu arteritis include autoimmunity and hypersensitivity response to proposed antigens, including heat shock protein and *Mycobacterium tuberculosis*. Histologically, granulomatous inflammation involves all layers of the vessel wall during the active phase of disease, followed by fibrotic stenosis.

Treatment generally starts with corticosteroids or other immunomodulatory therapy during the inflammatory phase of disease, followed by medical or interventional treatment to reduce organ injury.²⁴

Coarctation and Middle Aortic Syndrome

Coarctation occurs in about 1 in 1550 births and accounts for about one-third of all causes of RVH in infants. Interestingly, only 35% of isolated coarctation cases are diagnosed in the first year of life. Cases missed in childhood present as early onset hypertension in adults. Those cases treated in childhood can develop restenosis of the coarct segment later in life. It is estimated that 1 in 150 adults have congenital heart disease, with aortic coarctation accounting for 5% to 10%.

Adult presentations of RVH associated with coarctation include signs and symptoms of collateral development. Bruits may be heard over the carotids, and intercostal pulses may be palpable. A radial-femoral pulse delay is a sensitive physical examination finding. Present in only about 50% of cases, a harsh, systolic blowing murmur is heard best over the posterior thorax. Either MRA or CTA is necessary to confirm coarctation in adults. Indications for the treatment of coarctation in adults include upper limb hypertension and greater than 20 mm Hg systolic BP gradient across the stenosis with evidence of significant collateral flow. These patients have a fivefold excess risk of cerebral aneurysms, and screening cerebrovascular imaging is recommended. Current guidelines recommend multispecialty consultation among cardiologists, interventionalists, and surgeons to determine the optimal approach (endovascular vs. surgical) to repair. Success rates for cure of hypertension range from 69% to 80%, with highest chance for cure and best survival data when treated in children younger than 10 years. Adults with prior coarctation have a higher risk of developing hypertension than the general population. This risk is attributed to vascular noncompliance, reduced aortic arch baroreceptor sensitivity, early kidney injury with sustained RAAS activation, and, in some cases, restenosis of the repaired coarct segment.

A rare entity, middle aortic syndrome is a congenital segmental or diffuse narrowing of the abdominal or distal descending aorta, causing RVH usually noted in infancy. Concomitant proximal RA stenosis occurs in up to 80% of cases, with variants including RA atresia, hypoplasia, or dysplasia.²⁵ Associations between middle aortic syndrome and FMD, neurofibromatosis, Williams syndrome, and Takayasu arteritis have been reported, and these cases can present later in life. Middle aortic syndrome can cause claudication of the lower extremities and mesenteric ischemia. Angioplasty and stenting of stenotic segments, surgical bypass grafting, and autotransplantation of ischemic organs are treatment options.²⁵

Renal Artery Aneurysms

Renal artery aneurysms are a rare cause of RVH and can be associated with atherosclerosis, FMD, and vasculitis. Thrombosis within an aneurysm can lead to distal emboli and renal infarcts (Fig. 43.6). Aneurysms

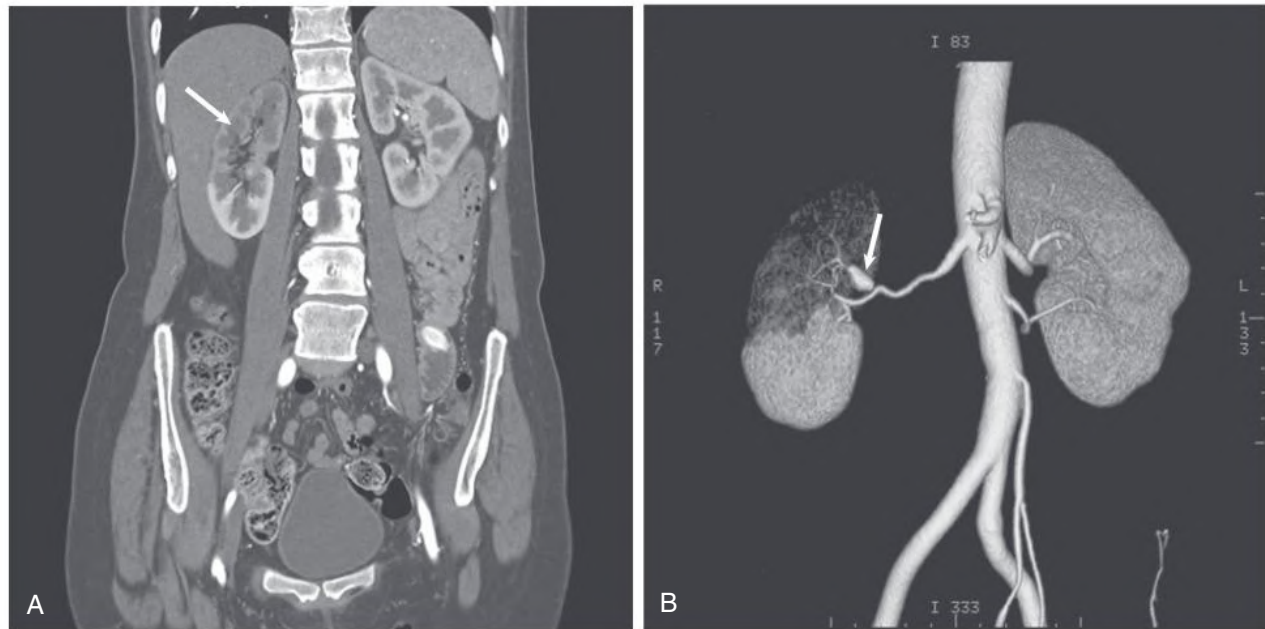


Fig. 43.6 Computed Tomography Angiogram of Renal Artery Aneurysm With Area of Infarction in Right Kidney. (A) Coronal image demonstrates area of intact tissue with no blood perfusion within the kidney parenchyma (arrow). (B) Reconstructed view with vascular aneurysm (arrow) and minimal flow in the distribution beyond this segment consistent with near-total occlusion. This patient presented with accelerated renovascular hypertension treated primarily with renin-angiotensin system blockade.

with diameters of greater than 1.5 cm have a significant risk of rupture. Other complications of RA aneurysms include vessel dissection and arteriovenous fistula formation.

ATHEROSCLEROTIC RENOVASCULAR DISEASE OR ISCHEMIC RENAL DISEASE

A second clinical presentation of renovascular disease is kidney dysfunction. Recent investigations have broadened our understanding of the processes leading to kidney injury, atrophy, and reduced glomerular filtration rate (GFR) in the setting of ARVD. Formerly called ischemic renal disease (IRD), it is now recognized that other factors in addition to ischemia are important in the pathogenesis. Severe RA stenosis or occlusion can decrease GFR by reducing glomerular capillary pressure below the threshold for renal autoregulatory compensation. This hemodynamic effect on kidney function occurs when renal blood flow is severely compromised. The histologic correlates of this reduced renal perfusion pressure include glomerular involution with basement membrane wrinkling and tubular cell atrophy and simplification and loss of brush border. The reduced energy-dependent tubular function and energy expenditure matches the reduction in oxygen delivery. Revascularization to improve renal blood flow can restore GFR and reverse these adaptive changes. In this scenario, the kidney is not chronically ischemic. Studies using blood oxygen level-dependent magnetic resonance imaging (BOLD MRI) indicate that, despite reductions in blood flow and GFR, many patients with RA stenosis maintain normal renal cortical and medullary tissue oxygenation.^{26,27} Thus, many poststenotic kidneys have no more ischemia than normal kidneys.

However, it is clear from randomized controlled trials evaluating the effects of revascularization on retrieval of kidney function, studies in experimental RA stenosis, and renal biopsies of poststenotic kidneys that factors other than hemodynamics play a significant role in the pathogenesis of ARVD (Fig. 43.7). Severe RA stenosis may cause

intermittent ischemic insults associated with acute or chronic regional or global hypoxia, which trigger maladaptive processes and ongoing kidney injury even when perfusion is restored.²⁸ These insults might be caused by fluctuations in BP and renal perfusion, embolic events, or progression of stenosis. Biopsies from poststenotic kidneys demonstrate influx of inflammatory cells and macrophages and increased expression of inflammatory cytokines including monocyte chemoattractant protein-1, tumor necrosis factor- α , interferon- γ , and interleukin-6. Biomarkers of tubular injury, including neutrophil gelatinase-associated lipocalin, are increased in poststenotic kidney RVs. Profibrotic factors including TGF- β and mitogen-activated protein kinase p38 are upregulated and associate with interstitial fibrosis and tubular atrophy. Ischemia leads to production of reactive oxygen species, which cause increased renal microcirculatory vascular tone and endothelial cell dysfunction.²⁹ This results in remodeling and rarefaction of the renal microvasculature at the level of the interlobar, arcuate, interlobular arteries and small arterioles.³⁰ This is associated with a decrease in expression of angiogenic factors including vascular endothelial growth factor (VEGF). Kidney hypoperfusion and ischemic-reperfusion insults also result in renal mitochondrial dysfunction.³¹

Through these processes, unilateral atherosclerotic RA stenosis leads to atrophy and dysfunction of the poststenotic kidney. The contralateral kidney often hypertrophies and compensates with hyperfiltration but develops progressive parenchymal injury mediated by the combined effects of high pressure and Ang II. In some cases, the kidney with the patent RA has worse function than the poststenotic kidney. In addition, studies in animal models of RA stenosis support the concept that the contralateral kidney participates in activation of proinflammatory and profibrotic pathways, which negatively affect the response of the poststenotic kidney to revascularization. In rats, nephrectomy of the contralateral kidney promotes recovery of the poststenotic kidney after reperfusion.³²

In summary, reduced renal perfusion and ischemia ultimately activate numerous mechanisms of tissue injury. This results in macrophage

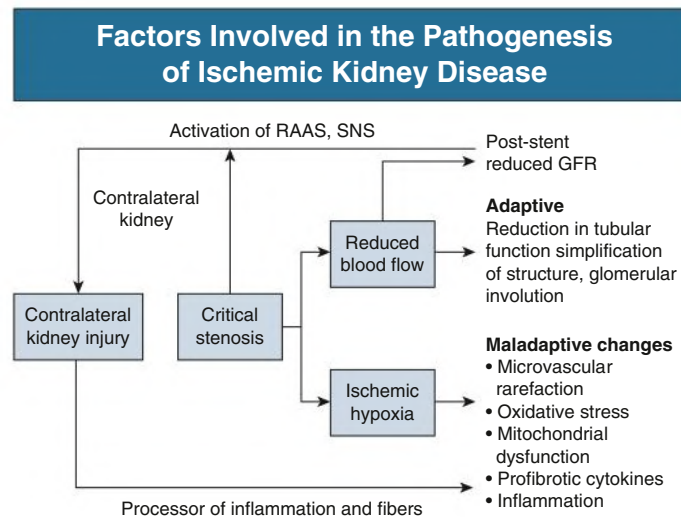


Fig. 43.7 The factors involved in the pathogenesis of ischemic renal disease (kidney dysfunction associated with atherosclerotic renal vascular disease). *GFR*, Glomerular filtration rate; *RAAS*, renin-angiotensin-aldosterone system; *SNS*, sympathetic nervous system.

accumulation with progressive tubular cell loss and interstitial fibrosis. Glomeruli are usually preserved but often appear involuted. The ischemic kidney develops microvascular rarefaction, contributing to ongoing irreversible structural and functional changes. Atherosclerotic renovascular disease or IRD comprises both the potentially reversible hemodynamic component and these adaptive and structural changes, some of which are not reversed by revascularizing the kidney.

Clinical presentations of ARVD or IRD are outlined in [Box 43.5](#). The most common is unexplained CKD in patients with generalized atherosclerosis. Often, BP is well controlled, particularly when RAAS blockade is used, though it may present with resistant hypertension. Risk factors for atherosclerosis are often present: long-standing hypertension, cigarette smoking, and hyperlipidemia. When investigated, these patients are found to have unilateral or bilateral RA stenosis and often have small or asymmetric kidneys. As an etiology of CKD, ARVD is usually associated with minimal proteinuria. Proteinuria develops in response to kidney injury over time and is a predictor of poorer response to revascularization. Urinary sediment is bland.

Atherosclerotic renovascular disease can also present as acute kidney injury (AKI) in patients with RA stenosis treated medically or in those with recurrent stenosis within a stented RA. New or episodic AKI, or AKI on CKD with recent rapid decline in eGFR in patients with known RA stenosis or those with diffuse atherosclerosis, should prompt evaluation for ARVD. In patients with high-grade RA stenosis, AKI can follow normalization of systemic BP with any agent. The sudden reduction in systemic BP can reduce RA pressure below levels needed to sustain GFR. With RAAS inhibitors, these alterations in glomerular hemodynamics may be more common or pronounced.³³ Normally, activation of Ang II causes efferent arteriolar vasoconstriction, which preserves transcapillary filtration pressures at the glomerulus when preglomerular pressures are reduced, thereby maintaining GFR. The loss of this compensatory mechanism induced by RAAS blockade can result in functional AKI. This typically occurs within a few days from the start of therapy and is usually reversible. An acceptable rise in serum creatinine above baseline is less than 0.5 mg/dL or a less than 30% change in estimated GFR (eGFR). If the change exceeds this, ARVD should be considered. An excessive drop in eGFR after initiation of RAAS inhibitors is not specific for the presence of RA stenosis and is seen frequently in patients with cardiac or hepatic

BOX 43.5 Clinical Presentations of Ischemic Renal Disease Associated With Atherosclerotic Renal Artery Stenosis

- AKI with control of BP: ACE inhibitors or ARBs
- AKI with aggressive diuresis in patients with congestive heart failure
- CKD otherwise unexplained in atherosclerotic age range
- CKD with asymmetric renal size
- Acute or chronic kidney disease and renovascular hypertension
- Acute on chronic kidney injury with episodes of “flash” pulmonary edema
- Unexplained rapid decline in GFR in CKD
- Oligoanuric kidney failure not otherwise explained in a patient with atherosclerosis and hypertension
- Renal atrophy

ACE, Angiotensin-converting enzyme; *AKI*, acute kidney injury; *ARB*, angiotensin receptor blocker; *BP*, blood pressure; *CKD*, chronic kidney disease; *GFR*, glomerular filtration rate.

dysfunction or patients with intravascular volume depletion because, in these settings, maintenance of GFR is also Ang II dependent.

Oligoanuria in the setting of AKI in patients with atherosclerosis or risk factors should prompt testing for critical renovascular disease. Often, concomitant RVH is present.

RENOVASCULAR DISEASE AND HEART FAILURE AND UNSTABLE CARDIAC CONDITIONS

Some patients with RA stenosis present with recurrent episodes of “flash” pulmonary edema, called Pickering syndrome.³⁴ These presentations are usually characterized by hypertensive urgency or emergency, hypervolemia, and echocardiographic evidence of diastolic dysfunction.³⁵ Such patients have increased mortality and hospitalization rates compared with those who have CHF without renovascular disease.³⁶ Case series and retrospective reviews suggest that renal revascularization can facilitate volume management, reduce hospitalizations, and improve GFR and cardiac function in this high-risk subgroup of patients with atherosclerotic RA stenosis.^{37,38} This cohort of patients has been underrepresented in randomized controlled trials evaluating the relative efficacy of revascularization and medical therapy. Some

believe that chronic stimulation of the RAAS secondary to RA stenosis contributes to the abnormal left ventricular remodeling in patients with chronic systolic heart failure, as well as to the frequency of CHF decompensation.³⁹

Patients with CHF have a high prevalence of concomitant RA stenosis. About half of patients with a diagnosis of systolic heart failure have concomitant RA stenosis and nearly 70% of those with both CHF and CKD.⁴⁰ The presentation of decompensated CHF, worsening kidney function, and hypertension should raise suspicion for this diagnosis along with other forms of cardiorenal syndrome. In cases with severely depressed left ventricular function, hypertension may be absent. Clinicians should consider the presence of critical RA stenosis if GFR continues to deteriorate during treatment of decompensated CHF.

Finally, some patients with RA stenosis and LVH can present with chest pain syndromes with no significant lesions in the coronary arteries.

IMAGING RENOVASCULAR HYPERTENSION AND RENAL ARTERY STENOSIS

Conventional direct angiography remains the reference standard to define the RA anatomy. Noninvasive options include RA duplex ultrasound, CTA, and MRA, each with their limitations and strengths.

Renal artery duplex scanning is often used to identify and follow hemodynamic effects of RA stenoses. It is most effective in detecting lesions of the main RA near the ostium and thus is a good screening test for atherosclerotic RA stenosis. The reliability of duplex ultrasound depends on the skill and experience of the operator and the body habitus of the patient. Duplex ultrasound provides little functional information beyond the vascular lesion, although structural features such as kidney size and echogenicity can be determined. The duplex diagnostic criteria for hemodynamically significant RA stenosis consider the comparative rates of blood flow in the stenotic area to that in the remaining segments of the RA and the aorta. Parameters measured include peak systolic velocity (PSV) at various sites along the RA and in the suprarenal aorta; the renal aortic ratio (RAR), which compares the PSVs at these segments; acceleration time and index, which help evaluate the RA waveform; and the intrarenal resistive index. The resistive index has been associated with intrinsic small vessel renal disease, and a value greater than 0.80 has a strong negative predictive value for a good BP response to intervention.⁴¹ Normal PSV in the RA will depend on the lumen diameter of the vessel and ranges from about 120 to 160 cm/sec. Peak systolic velocity greater than 200 to 220 cm/sec and RAR greater than 3.0 to 3.5 usually indicate RA stenosis of greater than 60% narrowing.⁴² Most laboratories provide interpretation as to whether parameters support at least 60% luminal diameter narrowing, but clinicians should be familiar with aortic and RA velocities and waveforms to evaluate the validity of these interpretations. Generally, the higher the PSV, the more severe is the stenosis. However, a very blunted (tardus-parvus) RA waveform can represent critical RA stenosis in the absence of an elevated PSV. Renal artery duplex allows for serial testing, hence monitoring for progression and changes in kidney size. It is the preferred test for evaluating for restenosis of a stented RA segment. Duplex screening can miss accessory renal vessels.

Three-dimensional MRA with gadolinium enhancement provides excellent visualization of the arteries and functional information about the kidneys (Fig. 43.8). Limitations include interobserver variability, a tendency to overestimate degree of stenosis, interference by motion and breathing artifact, and limited sensitivity for middle and distal vascular lesions and small accessory vessels. Magnetic resonance angiography has the advantage of avoiding radiation exposure. The use

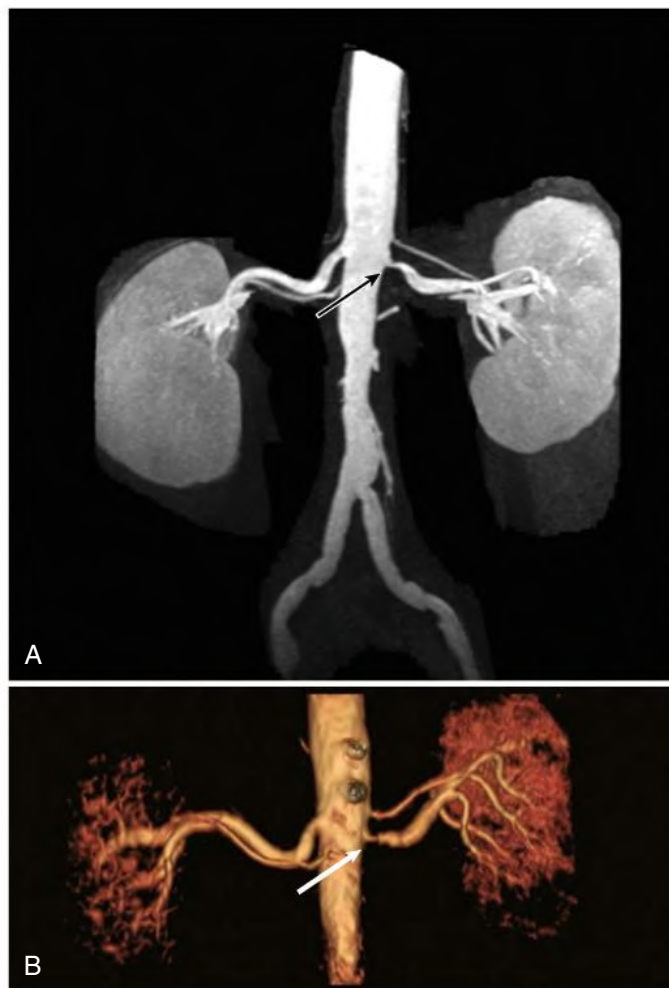


Fig. 43.8 Magnetic Resonance Angiogram With and Without Gadolinium Contrast Enhancement. (A) High-grade stenosis affecting the left inferior renal artery is evident, with functioning kidney tissue as reflected by gadolinium nephrogram (*black arrow*). Concerns about the role of gadolinium in the development of nephrogenic systemic fibrosis have greatly reduced the use of this contrast agent. (B) As a result, methods to image the vasculature without contrast material are being developed that produce excellent reconstructed images (*white arrow*).

of group II gadolinium-based contrast media is now considered safe for use in imaging in patients with AKI and advanced CKD or kidney failure with an extremely low incidence, if any, of nephrotoxicity or nephrogenic sclerosing dermatopathy.⁴³

In CTA with vascular reconstruction, imaging definition is nearly equal to that of conventional angiography but requires significantly more contrast, usually 100 to 125 mL. Focal vascular calcification often obscures accurate assessment of stenoses. Computed tomographic angiography is highly sensitive for identifying lesions in the mid and branch vessels often associated with FMD and is the recommended screening test for these patients based on recent consensus recommendations.⁸

Angiography remains the gold standard for defining the degree of stenosis and is the most reliable modality for identifying distal and branch or small vessel disease, which may be missed by other screening modalities. Aortography provides the opportunity to perform intravascular ultrasound to assess lesions and for the measurement pressure gradients across a stenosis, an aid in determining the hemodynamic significance of a lesion. When there is complete proximal arterial

occlusion, direct aortography can identify distal reconstitution by collaterals and a renal “blush” confirming parenchymal viability. Limited selective renal angiography can be performed with as little as 10 mL of contrast. In cases at highest risk for contrast-induced AKI, carbon dioxide can be used instead of nonionic iodinated contrast.

In cases of unilateral RA disease, captopril renography provides functional information regarding the size and excretory capacity of the kidney and demonstrates the rate of isotope appearance as an index of renal blood flow and filtration. It detects the presence of a differential role of Ang II on GFR between the kidneys. This test has a high negative predictive value for the presence of RVH when completely normal.⁴⁴ It is less specific in the setting of intrinsic kidney dysfunction.

Renal vein renin measurements may help predict the BP response to renal revascularization. Previous studies indicated that lateralization of RV levels (>1.5:1 stenotic-to-nonstenotic kidney ratio) predicts a favorable BP response for more than 90% of patients. Because failure to lateralize also carries a favorable response in almost half of patients, the negative predictive value is limited. Some clinicians use these measurements to verify lower renin levels in the pressor (i.e., nonstenotic) kidney before undertaking nephrectomy.

TREATMENT OF FIBROMUSCULAR DYSPLASIA AND ATHEROSCLEROTIC RENOVASCULAR DISEASE

Medical and Endovascular Treatments

Fibromuscular dysplasia of hemodynamic significance should be treated with revascularization. Current consensus defines significance as having a translesional pressure gradient of 10% of the mean aortic pressure. The treatment of choice for FMD is percutaneous renal artery angioplasty (PTRA). Stenting is not recommended and is reserved as a rescue procedure for complications of PTRA such as dissection. Successful PTRA results in disruption of the abnormal collagen bands in the lumen of the artery and the vascular wall leading to larger lumen diameter and less turbulent RA blood flow. Complete cure, defined as normalization of BP without the need for medications, occurs in 35% to 45% of cases. In more than 85% of cases treated with PTRA, BP is improved with reduction in the number of antihypertensive medications.⁴⁵ Predictors of response to intervention include lower preintervention systolic BP, younger age at treatment, shorter duration of hypertension, and a positive pretreatment captopril renogram.⁴⁶ Primary technical success rates exceed 90%. Inadequate initial treatment or restenosis has been reported in up to 34% of treated cases and is most common with multifocal variant. Subsequent procedures to address all areas of stenosis may be required. Use of intravascular ultrasound to guide endovascular treatment of these lesions can improve treatment success. When FMD is associated with aneurysmal dilations greater than 1.5 cm in diameter, surgical revascularization or endovascular exclusion using a covered stent may be required. Women of childbearing age with RA aneurysms should be treated before pursuing pregnancy because of the risk for rupture during pregnancy or delivery. Other recommended treatments for patients with FMD are antiplatelet agents and smoking cessation.

The optimal treatment of atherosclerotic RA stenosis presenting with RVH or other clinical syndromes is controversial. Primary percutaneous renal artery stenting (PTRS) became standard treatment in many centers during the mid to late 1990s. Target vessel patency rates using PTRS regularly exceed 95%. Between 1996 and 2005, enthusiastic application of PTRS in the treatment of atherosclerotic RA stenosis led to a marked rise in stent placement in these patients. Most case series and observational studies reported stabilization of BP and renal function in half of patients undergoing PTRS, improvement in up to 25%, and a decline in kidney function after PTRS in 25%.

The relative effectiveness of endovascular treatment of atherosclerotic RA stenosis compared with medical therapy has now been studied in several randomized controlled trials. These are summarized in Table 43.1. The two largest trials, the Angioplasty and Stent for Renal Artery Lesions (ASTRAL) and Cardiovascular Outcomes in Renal Atherosclerotic Lesions (CORAL), comparing medical therapy and stenting to medical therapy alone, demonstrated no differences in the primary endpoints between groups.^{47,48} In the ASTRAL trial, 806 subjects with hypertension, many with CKD, were randomized to PTRS versus medical therapy and followed for a mean of 33.6 months. Of the enrolled patients, 53% had bilateral RA stenosis; the mean degree of luminal narrowing was 75.5%. The ASTRAL trial reported no difference in the primary endpoint of change in kidney function as measured by reciprocal creatinine. Investigators in the CORAL trial enrolled 947 patients, all treated with well-defined medical therapy addressing all cardiovascular risk factors, used RAAS blockade as the cornerstone of antihypertensive therapy, and randomized half the group to medical therapy plus stent. The CORAL trial had an imaging oversight laboratory and allowed enrollment based on angiographic as well as CTA, MRA, and duplex ultrasound definition of RA stenosis greater than 60%. At a median of 43 months of follow-up, there were no differences in the composite endpoint of mortality, cardiovascular and renal events, or any of the individual components between the treatment groups.

Current consensus is that the initial treatment of patients with RVH and stable CKD associated with atherosclerotic RA stenosis should be a focused medical management approach to addressing all cardiovascular and renal risk factors in addition to hypertension. In CORAL, there was no difference in number of medications required to control BP between groups. For those patients with diabetes and CKD, attention to achieving hemoglobin A_{1C}, as well as CKD mineral bone disease and anemia targets, are important therapeutic goals. Statins are recommended in these patients both for cardiovascular risk reduction and with the goal of slowing progression of the RA lesion; there are also experimental data suggesting that statins attenuate kidney parenchymal injury associated with atherosclerotic renovascular disease.^{49,50}

Whether all patients with RA stenosis should be treated with RAAS blockade remains controversial. There is a risk for RAAS blockers to induce AKI, and this risk is higher in patients with bilateral RA stenosis or stenosis to a single functioning kidney. Although the unique properties of RAAS blockers allow more effective BP control in patients with RVH, there is the potential for early loss of filtration pressure in patients with critical RA stenosis.

Clinical experience with RAAS blockers in the treatment of RVH is reassuring. Registry data and prospective follow-up studies in patients with atherosclerotic RA stenosis indicate that blockade of the RAAS is usually well tolerated.^{51,52} In the CORAL trial, in which 20% of patients had bilateral RA stenosis, most patients received an angiotensin receptor blocker or angiotensin-converting enzyme (ACE) inhibitor. The lack of differences in kidney endpoints suggests that this was generally well tolerated. However, it is strongly advised that patients with atherosclerotic RA stenosis who are prescribed these agents have electrolytes and creatinine measured within 2 to 4 weeks after starting these agents and regularly over the course of follow-up.

Indications to Consider Kidney Revascularization in Atherosclerotic Renal Artery Stenosis

Despite the lack of randomized controlled data and the risks, some high-risk patients may benefit from RA revascularization when medical therapy falls short.⁵³ Indications to consider revascularization are listed in Box 43.6. First, uncontrolled hypertension, despite all efforts

TABLE 43.1 Randomized Controlled Trials Comparing Medical Therapy to Medical Therapy With Renal Artery Stenting for Renal Artery Stenosis

Study	ASTRAL (2009)	STAR (2009)	CORAL (2014)
Cohort	Hypertension	Hypertension and CKD	Hypertension and/or CKD
Entry BP	No BP threshold required	BP <140/90 mm Hg and stable for 1 month and eGFR <80 mL/min	SBP >155 mm Hg on two or more medications or eGFR <60 mL/min
Stenosis	>50% by MRA, CTA, angiography	>50% by MRA, CTA, or angiography	>60% by MRA, CTA, angiography, DUS
Excluded	Clinician certain patient would benefit from stent or require stent within 6 months	Malignant hypertension Pulmonary edema with bilateral RA stenting Intolerance to ACEI/ARBs as evidenced by >20% drop in CrCl	Entry creatinine >4 mg/dL Kidney length <7 cm
% Stenosis	75.5 mean %	NA	67.3%/66.2%
CKD	Mean creatinine 2.0 mg/dL	Mean creatinine 1.7 mg/dL	Mean eGFR 58 mL/min/1.73 m ²
% Bilateral	53.5%	47.9%	22%
Subjects per arm (N/N)	403/403	76/64	459/472
Follow-up	33.6 months	24 months	43 months
Treatment	Stent	Stent	Stent
Medical treatment	At discretion of sites BP control with or without ACEI or ARB No specified target BP	BP target <140/90 mm Hg ACEI/ARB last resort ASA Statin Smoking cessation counseling	BP target <140/90 mm Hg 130/80 mm Hg for DM and CKD ACE/ARB first-line ASA Statin goal LDL <70 mg/dL Hb A _{1c} <7.0% for DM Smoking cessation counseling
Endpoint	Rate of progression of CKD based on reciprocal creatinine over time	≥20% decrease in CrCl	Composite cardiovascular and renal events
Outcome	No significant difference	No significant difference	No significant difference

ACEI, Angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; ASA, aspirin; ASTRAL, Angioplasty and Stent for Renal Artery Lesions; BP, blood pressure; CKD, chronic kidney disease; CORAL, Cardiovascular Outcomes in Renal Atherosclerotic Lesions; CrCl, creatinine clearance; CTA, computed tomography angiography; DM, diabetes mellitus; DUS, Doppler ultrasound; eGFR, estimated glomerular filtration rate; HbA_{1c}, hemoglobin A_{1c}; LDL, low-density lipoprotein; MRA, magnetic resonance angiography; NA, not applicable; N/N, number of subjects in each arm; RA, renal artery; SBP, systolic blood pressure; STAR, Stent Placement in Patients with Atherosclerotic Renal Artery Stenosis and Impaired Renal Function.

Data from Postma CT, van Oijen AH, Barentsz JO, et al. The value of tests predicting renovascular hypertension in patients with renal artery stenosis treated by angioplasty. *Arch Intern Med.* 1991;151:1531–1535; Ritchie J, Green D, Chrysochou C, et al. High-risk clinical presentations in atherosclerotic renovascular disease: prognosis and response to renal artery revascularization. *Am J Kidney Dis.* 2014;63(2):186–197; and Weinreb J, Rodby RA, Yee J, et al. Use of intravenous gadolinium-based contrast media in patients with kidney disease: consensus statements from the American College of Radiology and the National Kidney Foundation. *Radiology.* 2021;298:28–35.

BOX 43.6 Clinical Indications to Consider Renal Artery Revascularization

- Worsening kidney function due to IRD
- Uncontrolled hypertension failing medical therapy
- Intolerance to medical therapy
- Recurrent hospitalizations for pulmonary edema without other obvious cause while on optimal medical therapy
- Other unstable cardiac or renal trajectories
- Progressive renal atrophy (controversial)
- Potentially reversible dialysis dependence due to IRD

IRD, Ischemic renal disease.

to optimize pharmacologic and dietary interventions and to enhance adherence, should prompt evaluation for all potential causes of resistance. Blockade of the RAAS should be included in the antihypertensive regimen in these resistant cases. Some patients will be intolerant of the very medications they need to control BP. Others may present with

hypertensive urgency or emergency despite therapy. Some of these patients may respond to PTRS. Second, when RA stenosis critically reduces glomerular capillary pressure such that there is a rapid decline in GFR, renal revascularization may improve kidney function and avoid the need for dialysis. In these cases, there are often multiple other potential contributors to declining GFR that must be considered. Given the need to balance the potential risks and benefits of intervention, input from a multidisciplinary team including nephrologists, vascular surgeons, and interventionalists is often helpful. In one series, PTRS in dialysis-dependent patients with ARVD led to enough improvement in kidney function to allow for discontinuation of dialysis.⁵⁴ Finally, some patients with recurrent hospitalizations for decompensated heart failure or flash pulmonary edema that is attributed to medically resistant RA stenosis may benefit from RA revascularization.⁵³

Fig. 43.9 outlines a proposed clinical algorithm for managing patients with atherosclerotic RA stenosis.

For patients who undertake PTRS, surveillance for in-stent RA stenosis should be undertaken based on changes in either control of BP or kidney

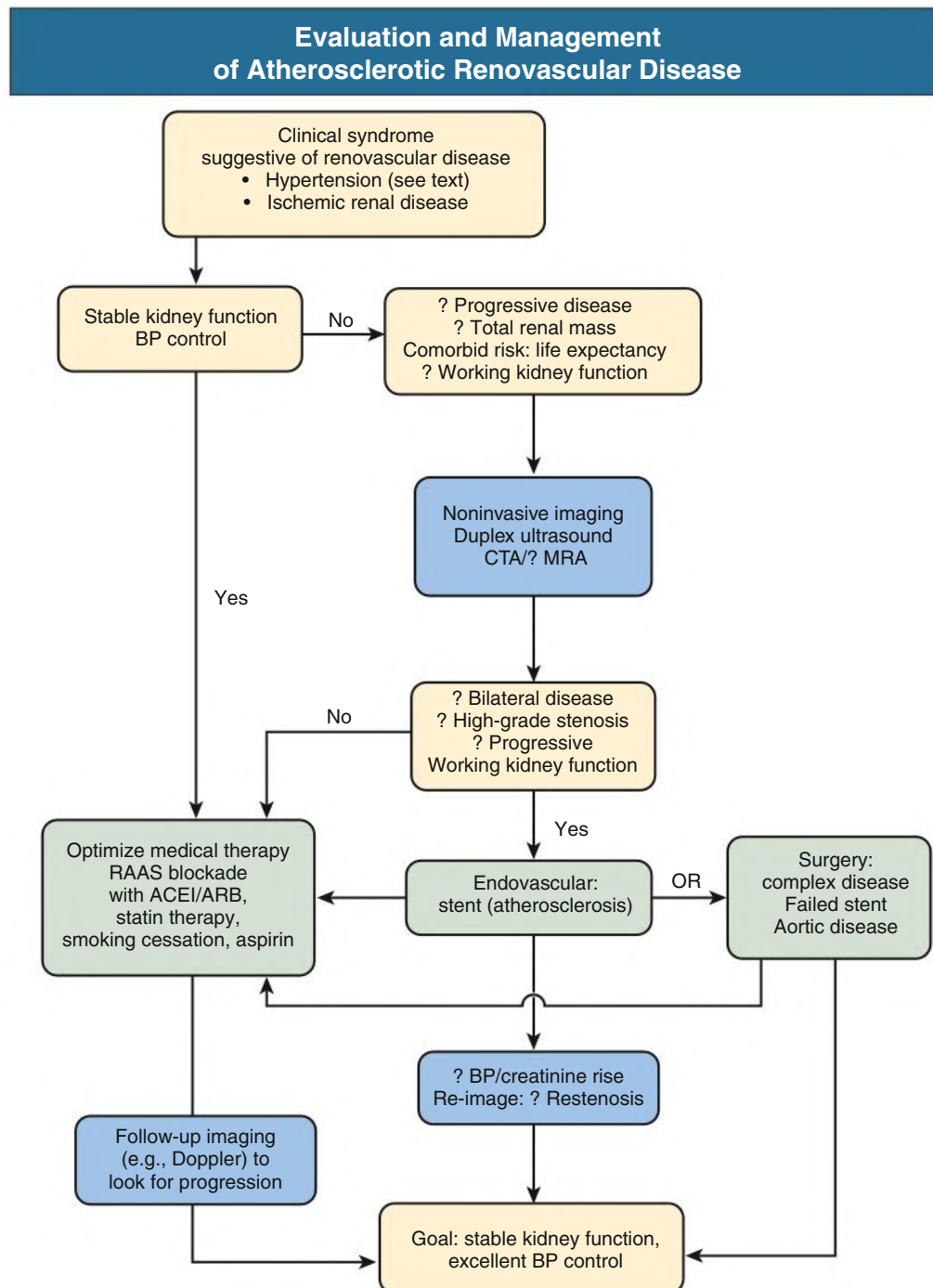


Fig. 43.9 Algorithm for the Management of Renal Artery Stenosis. The intensity of imaging and revascularization depends on both the level of kidney function and the blood pressure (BP), in addition to the comorbid disease risks for the individual patient. The overall goal should focus on stable kidney function and BP levels. As with any other vascular disease, monitoring for disease progression and recurrence is an important element of long-term management. *ACEI*, Angiotensin-converting enzyme inhibitor; *ARB*, angiotensin receptor blocker; *CTA*, computed tomographic angiography; *MRA*, magnetic resonance angiography; *RAAS*, renin-angiotensin-aldosterone system.

function. In-stent stenosis remains a significant complication occurring in up to 33% of cases and can occur at any time after stent placement, with the highest risk within the first year after primary PTRS.⁵⁵ Patients with RA stents should be followed closely at regular intervals for this complication, which can be treated to prevent occlusion. An example of stent fracture as a cause of RA thrombosis and infarction is demonstrated in Fig. 43.10.

Clinicians should inform patients with atherosclerotic RA stenosis about the risk and uncertain benefits of PTRS.

Surgical Renal Revascularization

Before the introduction of endovascular techniques, surgical revascularization was the standard treatment for patients with ARVD and



Fig. 43.10 Thrombosis of Renal Artery Complicating Renal Artery Stenting. Right and left renal artery stents. The left renal stent (arrow) is triangular, indicating crimping of the proximal portion, which in this patient was associated with thrombosis of the renal artery, seen here as no contrast entering the vessel. The right renal stent is patent.

RVH. Such procedures carry considerable risk, cost, and morbidity. As a result, surgical intervention for renovascular disease is generally reserved for patients who do not respond to best medical and endovascular therapy or who have associated aortic disease that is not amenable to endovascular therapy. Despite these caveats, successful surgical revascularization in well-selected cases provides durable restoration of kidney blood supply with good long-term patient survival.⁵⁶ Overall, the effects of surgical revascularization on BP and renal function response in patients with atherosclerotic RA stenosis mirror those for PTRS. Surgical revascularization in the modern era may provide more durable patency than PTRS with lower complication rates and risk for restenosis.

Surgical revascularization should be considered in patients with total occlusion of the RA and abrupt loss of GFR for retrieval of kidney function. Some dialysis-dependent patients and some with advanced CKD with ARVD experience recovery of kidney function after revascularization.⁵⁷ Treatment involves assessment of the risks versus potential benefits of heroic revascularization procedures. The status of the contralateral kidney and overall residual kidney function should be weighed against the potential retrievable function from the underperfused kidney, as well as the perioperative risk associated with surgery. Predictors of GFR recovery with revascularization in the setting of critical RA occlusion include preserved kidney size, evidence of a renal blush or nephrogram by imaging, recent decline in GFR, and recent baseline creatinine concentration less than 3 mg/dL.⁵⁸

When a poorly functioning atrophic kidney causes RVH refractory to optimal medical therapy including RAAS blockade, nephrectomy of this pressor kidney may improve BP control with minimal impact on total GFR.

Atherosclerotic Renal Vascular Disease: Beyond the Stenosis

The recognition that ARVD or IRD is more than a hemodynamic disorder has led to early investigation into targeting the inflammatory ischemic-reperfusion and vascular rarefaction aspects of this form of kidney disease. Early studies suggest that intraarterial infusion of autologous mesenchymal stem cells increase tissue oxygenation, blood flow, and GFR in poststenotic kidneys, with associated reduction in

inflammatory biomarkers. Other studies looking at administering VEGF and strategies to mitigate mitochondrial dysfunction in models of ARVD have also shown promising results.⁵⁹

Renovascular Disease in the Renal Transplant

Recognition of vascular complications of renal transplantation can prolong allograft survival. Early posttransplant, RA, and vein stenosis or thrombosis are rare but treatable complications.

Renal artery thrombosis occurs in about 0.4% of transplants within minutes to hours after transplantation. Causes include hyperacute rejection, kinking of the RA, anastomotic arterial occlusion, or intimal flap. It has also been reported associated with decubitus positioning.

Transplant RV thrombosis is more common, occurring in 5% of adult transplants and 8.2% of pediatric cases, usually within the first 5 days after transplantation, and is more common with right kidney donation. It usually leads to graft infarction, but rupture of the allograft can occur. Risk factors include donor age over 60 years, donor artery atherosclerosis, and longer allograft cold ischemia time. Recipient risk factors include age younger than 6 years or older than 50 years, hypovolemia in the perioperative period, multiple RVs, DM, technical or surgical complications, and prior peritoneal dialysis. When thrombotic vascular complications occur without obvious cause, an evaluation for hypercoagulable state is recommended. There are some data supporting the protective effects of low-dose aspirin in this population. Renal salvage is possible with early diagnosis, surgical exploration, and thrombectomy.

The most common posttransplant vascular complication is transplant RA stenosis, which occurs between 3 months and 2 years posttransplantation and accounts for about 5% of all posttransplant hypertension. The incidence ranges from 1.3% to 23%, depending on the screening tests used.⁶⁰ In many cases, anastomotic stenoses are not hemodynamically significant. Renal artery stenosis is more common in deceased donor transplants than in living donor transplants and in allografts with multiple renal vessels. The use of pediatric kidneys in adult recipients is associated with a higher rate of stenosis because of greater turbulent flow from mismatch between donor and recipient vessel size.

The cause for transplant RA stenosis is usually intimal scarring and hyperplasia in response to trauma to the vessel during harvesting or anastomotic stenosis, which is most commonly associated with end-to-end anastomoses and may be related to suture technique. In end-to-side anastomoses, stenosis is typically postanastomotic, suggesting that turbulence or other hemodynamic factors play a role. Immunologic causes of transplant RA stenosis have been proposed on the basis of histologic similarities with chronic vascular rejection and association with prior acute rejection. In addition, de novo class II donor-specific antibodies have been associated with postanastomotic stenosis.⁶¹ Other possible pathogenic mechanisms include atherosclerosis of the donor artery, calcineurin inhibitor toxicity, and cytomegalovirus infection. External compression on the transplant artery by large native polycystic kidneys or pseudoaneurysms have also been reported causes. Renal artery stenosis occurring many years after transplantation most often represents atherosclerotic disease. As the transplant population ages, there has been increasing recognition of another subset of patients with pseudo-transplant RA stenosis, in which atherosclerotic vascular disease proximal to the allograft artery, particularly involving the iliac vessel, results in reduced kidney perfusion.

Patients typically present with new-onset or worsening hypertension with or without kidney dysfunction. An example of the 1-clip-1-kidney Goldblatt model, transplant RA stenosis often presents with fluid retention and even flash pulmonary edema. Often, a two-component RA bruit can be heard over the allograft.

Pseudo-transplant RA stenosis, which is suprarenal iliac artery stenosis, often presents with ipsilateral lower extremity claudication along with hypertension and worsening allograft function. Risk factors for the development of transplant RA stenosis include older donor age and extended criteria donor and recipient male sex, DM, hyperlipidemia, smoking, ischemic heart disease, and elevated serum creatinine at discharge from transplantation.⁶²

Renal duplex ultrasound is the initial screening test for transplant RA vascular complications, with the sensitivity and specificity ranging from 90% to 100% and 67% to 100%, respectively, depending on parameters and cut-off values. Compared with native renal arteries, data show that PSVs are often higher at the anastomosis even without stenosis due to the angle at the anastomosis, with 26% of normal transplant RAs having a PSV greater than 250 cm/sec at 9 months posttransplant.⁶³ Using a combination of a PSV greater than 300 cm/sec, acceleration time of more than 0.1 second, and the presence of poststenotic spectral broadening (indicative of turbulent flow), the probability of stenosis approaches 99%.⁶⁴ Magnetic resonance angiography provides excellent anatomic definition of the transplant RA. It may be associated with clip artifact at the anastomosis. Non-contrast-enhanced MRA has emerged as a good option for visualizing the iliac artery and the transplant arteries.⁶⁵ Computed tomographic angiography is comparable to renal arteriography but requires more contrast. Carbon dioxide angiography can allow for visualization and successful treatment using as little as 9 mL of iodinated contrast.⁶⁶

Percutaneous renal artery angioplasty with or without stenting is often the initial approach to transplant RA stenosis, with success rates of 90% for technical outcomes and 66% to 94% for clinical outcomes (BP and kidney function improvement) and a complication rate of 9.9% including vessel dissection, thrombosis, and site hematoma.⁶⁷ Restenosis rates are higher after primary PTR (18.9%) versus stenting (9.1%). Long-term outcomes for transplant patients treated for transplant RA stenosis are reported to be equal to those without this complication.⁶⁸ Surgical renal revascularization of allografts is difficult and associated with high complication rates. Extensive fibrosis develops around the allograft and often involves the renal vessels, making surgical access to the renal vessels risky. Historically, complications include graft loss (in 15%–30% of cases), ureteral injury (14%), and death (5%). A recent small study of 10 cases requiring surgery for transplant RA stenosis or occlusion reported better patient and long-term graft outcomes.⁶⁹

KIDNEY INFARCTION

Kidney infarction represents a distinct clinical syndrome associated with abrupt loss of blood flow to kidney parenchyma. Small areas of the cortex or medulla or the entire kidney may be affected. Autopsy series suggest the incidence is between 0.5% and 1.5%, with incidences reported from emergency department series at 0.004% to 0.007%, indicating that this diagnosis is often missed.⁷⁰ Bilateral renal infarction is reported in up to 20% of cases. The clinical presentation of kidney infarction mimics that of more common disorders, such as kidney stones, pyelonephritis, and muscle strains. A high clinical suspicion is needed to pursue the diagnosis.

Risk factors associated with kidney infarction include preexisting hypertension, cigarette smoking, arrhythmias (especially atrial fibrillation), hyperlipidemia, structural heart disease, prior embolic events, and diabetes.

The most common symptom is loin, flank, or abdominal pain, occurring in over 95% of cases. Transient RVH occurs in about 50% of cases. Nausea and/or vomiting are present in up to 45% of cases. However, nearly 25% of cases are asymptomatic, identified only by

BOX 43.7 Causes of Renal Infarction

Renal Artery Thrombosis

Spontaneous

- Renal artery atherosclerosis or fibromuscular dysplasia
- Renal artery or aortic dissection
- Renal or aortic aneurysms

Traumatic

- Postprocedure
- Endovascular stents
- Renal transplantation
- Hypercoagulable disorders
- Malignancy
- Antiphospholipid syndrome
- Renal artery vasculitis
- Vascular rejection
- Thrombotic microangiopathies
- Genetic
- Ehlers-Danlos vascular variant

Renal Embolism

- Atrial fibrillation
- Cardiac mural thrombi
- Valvular heart disease
- Paradoxical embolism
- Tumor or fat embolism
- Atheroemboli

Renal Vein Thrombosis

- Nephrotic syndrome
- Post-kidney transplantation
- Traumatic

enhancement or functional defects on kidney imaging. Laboratory abnormalities include elevated lactate dehydrogenase levels in over 90% of cases, with leukocytosis and elevated inflammatory markers such as C-reactive protein in up to 78% of cases, and may last up to 30 days from presentation. Microhematuria and proteinuria are present in 20% to 60% of cases.⁷¹

Acute kidney injury frequently accompanies kidney infarction, reported in between 11.1% and 64.7% of cases. When bilateral occlusion of the RAs or infarction of a single functioning kidney occurs, the patient may present with oliguric or anuric AKI. The development of CKD after kidney infarction has been reported in between 6% and 44% of cases.⁷²

Computed tomography with intravenous contrast is the imaging modality of choice for the diagnosis of renal infarction. Findings include focal wedge-shaped areas of decreased attenuation or global infarction with or without a rim sign indicating intact collateral circulation. Perinephric stranding is common. Simultaneous infarcts in the liver and spleen occur in 10% to 15% of cases. Other potential imaging techniques include MRA with gadolinium or nuclear scintigraphy with dimercaptosuccinic acid.

Causes of kidney infarction are listed in Box 43.7. Common causes include RA embolism from a cardiac source, arterial injury from dissection, trauma, or complications of endovascular procedures, and hypercoagulable or prothrombotic factors. In approximately 18.4% of cases, no cause is found.⁷³ In a review of over 186 patients with renal infarction, a RA lesion (atherosclerotic, dissection, hematoma, or FMD) was the cause in over 81.2% of cases, with emboli accounting

for only 9.1%.⁷⁴ Spontaneous RA thrombosis or dissection can lead to RA occlusion. Spontaneous RA dissection has a male predominance, with a 4:1 male-to-female ratio and average age of onset in the mid 40s.

Renal infarction secondary to traumatic RA injury occurs in 1% to 4% of all nonpenetrating abdominal trauma associated with deceleration injury and direct blunt trauma to the loin or flank regions. Evidence of lumbar vertebral injury should raise suspicion in the emergency department for renovascular trauma. Traumatic renal vascular occlusion often leads to kidney infarction within 3 to 6 hours. The success rate for revascularization of traumatic RA thrombosis remains low, even when diagnosed early.⁷⁵

Emboli to the right and left RA occur with equal frequency, with 12% of cases being bilateral. Atrial fibrillation, cardiac thrombus after myocardial infarction, atrial myxoma or other cardiac tumors, endocarditis, paradoxical emboli, and aortic thrombus are the most common conditions associated with embolic kidney infarction. Atrial fibrillation is the most common cardiac cause with the highest risk during the first year after the onset and when anticoagulation is subtherapeutic.⁷⁶ When echocardiography is performed, cardiac thrombus is rarely detected. There is a 30-day mortality rate of 10% to 13% after renal infarction, due to atrial fibrillation. Other sources of emboli to the kidneys include fiber related to cardiac bypass, gel foam due to embolization procedures, calcium from valve annuli, and even “bullet emboli” in the setting of trauma. Paradoxical RA embolism may occur in patients with right-to-left cardiac shunts, most commonly due to atrial septal defects present in 9% to 35% of the general population. The diagnosis requires clinical, angiographic, or pathologic evidence of systemic embolization and an abnormal communication between the right and left circulations with a favorable pressure gradient (typically diagnosed by “bubble” echocardiography) for the passage of a clot from the right to the left side of the heart. The presence of source of venous thrombus should be sought.

Less common causes of infarction include hypercoagulable states, inflammatory diseases of the retroperitoneum, thrombotic microangiopathies, and RA vasculitis or vasospasm. Antiphospholipid antibody syndrome is associated with both arterial and venous thrombotic events and is the most common cause of spontaneous arterial thrombosis.⁷⁷ Rare causes of kidney infarction include autoimmune diseases and drug abuse, such as intravenous injection or nasal insufflation of cocaine or even marijuana smoking. During the COVID-19 pandemic, several reports of kidney infarction due to COVID-19-associated hypercoagulability emerged.⁷⁸

Aortic dissection can compromise renal blood flow and cause kidney infarction. Aortic dissection occurs most commonly in association with atheromatous vascular disease of the thoracic aorta, but it can occur in collagen disorders, such as Ehlers-Danlos type IV or Marfan syndrome, and with arteritis, such as Takayasu arteritis.

Kidney infarction can be a complication of endovascular procedures. Endovascular aortic stents commonly used to treat abdominal aortic aneurysms can compromise flow to main or accessory RA orifices.^{79,80} Increasingly, fenestrated or branched covered stent grafts have been designed to allow for preservation of branch visceral vessels. Accessory RAs supplying less than half of the kidney may be intentionally sacrificed during these procedures in up to 25% of cases. When this occurs, up to 75% of patients will experience renal infarction postprocedure. In some reports, this is associated with significant decline in eGFR at 3 years.⁸¹ Furthermore, RA stents are usually placed through fenestrations with a goal of preserving RA flow. Renal artery occlusion can complicate these procedures with development of occlusion or stenosis in up to 30% of cases by 4 years postprocedure. The aortic graft and branch stents alter the natural RA motion during the cardiac cycle, and kinetic forces over time can result in RA stent fracture or kinking,

leading to vessel occlusion. Aortic stent graft migration and endoleaks can also impair RA inflow.⁸²

Treatment of kidney infarction is usually conservative and includes pain control and treatment of associated RVH. If RA occlusion is caused by thrombosis associated with a hypercoagulable state or an embolism from a central source, systemic anticoagulation is indicated, and consideration should be given to thrombolytic therapy. Anticoagulation for idiopathic renal infarction is controversial. Echocardiography is indicated to seek intracardiac or aortic arch thrombi and valvular abnormalities. Emergent endovascular recanalization of thrombosed stents associated with fenestrated aortic grafts can sometimes be successful in averting global kidney infarction.

ATHEROEMBOLIC KIDNEY DISEASE

Atheroembolic kidney disease describes a subacute kidney injury due to occlusion of small arteries and arterioles by cholesterol emboli. It is estimated to account for up to 10% of unexplained kidney failure in the elderly, with 7% of renal biopsies in a large study of patients with AKI over 60 years of age showing histologic findings of atheroembolism.⁸³ Atheroembolism may occur in up to 30% of patients with extensive aortic atherosclerosis after endovascular interventions. The incidence of atheroembolism after cardiac catheterization is estimated near 1.4%.⁸⁴ Furthermore, studies using filters to capture embolic material confirm that PTRAs and PTRS release thousands of atheromatous particles of various sizes in 70% to 100% of cases.⁸⁵ Preprocedural treatment with antiplatelet agents and intraprocedural use of embolic protection devices may reduce the burden of embolic atheromata.⁸⁶ However, use of distal protection devices can be particularly challenging in the RAs due to shorter length and bifurcations.

Risk factors for atheroembolic disease include older age, male sex, cigarette smoking, diabetes, hypertension, and hyperlipidemia. Atheroembolic disease is iatrogenic in 70% of cases, that is, associated with arterial intervention such as angiography, cardiovascular surgery, thrombolytic therapy, or anticoagulation. Spontaneous atheroembolism may occur in patients with extensive atherosclerosis and unstable plaque.

Acute or subacute reduction in eGFR caused by kidney microinfarctions developing as long as 6 months after the atheroembolic insult is the most common presentation leading to the diagnosis. It is important to differentiate atheroembolic from thromboembolic disease. Atheroembolic particles originate from upstream unstable atherosclerotic plaque and cause downstream partial occlusion of small vessels. The cholesterol cleft induces a foreign body reaction leading to an interstitial inflammatory response and subacute kidney injury.

The clinical picture is multisystemic and involves the kidneys in about 75% of patients. If a large atheroembolic shower induces significant renal tubular damage, the resulting AKI may manifest with an oliguric phase characterized by a high fractional excretion of sodium. More often, the kidney failure is nonoliguric, slowly progressive, often with a stair-stepping subacute course. Some patients will have only moderate impairment in GFR, whereas others progress to kidney failure.⁸⁷ Urinalysis findings are nonspecific but may include mild proteinuria, microhematuria, pyuria, and eosinophiluria. Renin release by ischemic zones in areas of embolization can lead to labile hypertension early in the course, sometimes associated with transient marked proteinuria. Fever, often low grade, is characteristic.

Although the kidneys are the organs most commonly involved, extrarenal cholesterol embolization may provide clues to aid in the diagnosis. Eosinophilia and hypocomplementemia commonly occur in the acute phase, reflecting immunologic activation.⁸⁸ Cutaneous findings are seen in up to 60% of patients during initial presentation.



Fig. 43.11 Livedo Reticularis. The mottled skin changes associated with peripheral cholesterol embolization may be seen over the legs, buttocks, back, or flank and may be transient.

These findings include blue or purple toes, mottled serpiginous rash (livedo reticularis; Fig. 43.11), petechiae, and purpura or necrotic ulceration in areas of skin embolization, such as the lower back, buttocks, lower abdomen, legs, feet, or digits.

Other organs often involved include the spleen (55% of cases), pancreas (52%), gastrointestinal tract (31%), liver (17%), and brain (14%). Symptoms associated with extrarenal atheroembolic disease include abdominal or muscle pain, nausea, vomiting, ileus, gastrointestinal bleeding, ischemic bowel, hepatitis, angina, Hollenhorst plaques, and visual and neurologic deficits.

Atheroembolic kidney disease should be suspected when subacute kidney failure develops after a vascular intervention in the presence of livedo reticularis. Many laboratory abnormalities indicative of tissue injury are often present, including elevated erythrocyte sedimentation rate (97% of cases), elevated serum amylase (60%), leukocytosis (57%), anemia (46%), hypocomplementemia (especially low C3) (25%–70%), and elevated lactate dehydrogenase and creatine kinase (38%–60%). Peripheral eosinophilia, which may be transient, is seen in up to 57% of patients. The presence of eosinophilia should raise suspicion for atheroembolic renal disease in the appropriate clinical setting. Definitive diagnosis is made by biopsy of an involved organ. A skin or muscle biopsy in an involved area may preclude the need for kidney biopsy. However, tissue biopsy is usually unnecessary when the classical clinical triad is present: precipitating event, subacute kidney injury, and typical skin findings.

If clinical or other pathologic evidence has not secured the diagnosis, kidney biopsy may be helpful. Diagnosis is based on the presence of birefringent, biconvex, elongated cholesterol crystals or biconcave clefts within the lumina of small vessels left behind in formalin-fixed tissue (Fig. 43.12). Due to the patchy nature of this disorder, open-wedge kidney biopsy has a higher likelihood of successful diagnosis relative to a percutaneous approach because it allows visualization and

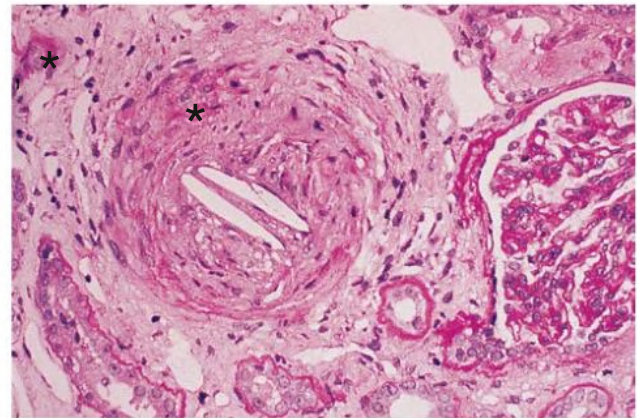


Fig. 43.12 Cholesterol Emboli in Kidney Biopsy Specimen. Biconvex cholesterol clefts with giant cell reaction (asterisks) and recanalization of the lumen of a medium-sized renal vessel. (Periodic acid–Schiff stain.) (Courtesy Dr. R. Horn, Vanderbilt University, Nashville, TN.)

direct sampling of areas of mottled cortex. In frozen sections of tissue, the cholesterol material can be identified with polarized light microscopy. The pathologic findings also may include intimal thickening and concentric fibrosis of vessels, giant cell reaction to the cholesterol particles, vascular recanalization, endothelial proliferation, tubulointerstitial fibrosis with both eosinophil and mononuclear cell infiltrates, glomerular ischemia, and even focal segmental glomerulosclerosis. In the kidney, the most commonly affected vessels are the arcuate and interlobular arterioles, leading to patchy ischemic changes distal to these vessels.

Atheroembolic kidney disease mimics other systemic disorders including vasculitis, infection, or thrombotic microangiopathy. Often these patients have undertaken procedures in which the differential diagnosis would include contrast nephropathy, acute tubular necrosis, renal infarction, or renal infarction. Peripheral eosinophilia and, in some cases, eosinophiluria, rash, fever, and reduced eGFR can be seen associated with acute interstitial nephritis. Chronic Cholesterol embolization syndrome may appear similar to hypertensive nephrosclerosis or IRD. In the kidney transplant recipient, kidney atheroembolism may mimic acute rejection or chronic allograft nephropathy.

The natural history is determined by the extent of organ involvement and the degree of the embolization. In one series of cases, eGFR declined rapidly in 29%, with a slower progressive course seen in 61%.⁸⁷ Among the latter group, the decline in eGFR was thought to result from a combination of cholesterol embolization and ARVD. Patients also may manifest acute or subacute reductions in eGFR followed by partial recovery. Conversely, the outcome can be dismal, particularly when cerebral embolization occurs or when there is a large unstable atheromatous burden. Some patients with cholesterol embolization may develop kidney failure.⁸⁹

There is no specific therapy for atheroembolic disease; treatment is supportive and focused on secondary prevention. As such, statins, aspirin, BP control, smoking cessation, and glycemic control are helpful measures in the prevention of atherosclerosis.

Prevention remains the most effective management strategy. Patients with extensive aortic atherosclerosis should be considered for alternative approaches to cardiac catheterization, for example, such as through the brachial artery. Once the diagnosis of cholesterol embolization has been established, further endovascular interventions should be avoided. When clinical factors dictate the need for aortic, renal, or peripheral arterial surgery, optimal timing and surgical approach are critical. Conversely, there is a growing surgical experience with segmental aortic replacement to remove the source of emboli, particularly

when atheroembolic disease occurs spontaneously. Transesophageal echocardiography is often used to identify mobile ulcerative plaque in the aorta to guide intervention.

Angiotensin-converting enzyme inhibitors are effective in managing the labile hypertension seen early in the course. Corticosteroids have been used with some success in patients with atheroembolic disease and associated inflammatory symptoms.⁹⁰ Several reports note stabilization of skin signs of cholesterol embolization after administration of statins.⁹¹ Because anticoagulation has been associated with atheroembolic disease, it is recommended to avoid anticoagulation in the acute phase. Although direct causality between anticoagulants and cholesterol embolization has not been established, the proposed mechanism is that anticoagulants prevent thrombus organization over the ulcerative plaques.

RENAL VEIN THROMBOSIS

Renal vein thrombosis is rare in adults most often associated with nephrotic syndrome, kidney tumors, hypercoagulable states, and after surgery or trauma to the renal vessels. When it occurs, the diagnosis is often never considered.

Thrombosis of the longer left RV also may involve the ureteral, gonadal, adrenal, and phrenic branches that drain into the left vein, whereas on the right side, the adrenal and gonadal veins drain directly into the inferior vena cava. Because of the larger network of venous complexes, occlusion of the left RV results in enlargement of the systemic collateral vessels, which provides some protection against infarction, which is a rare complication of RV thrombosis on either side.

Experimentally, acute RV thrombosis is associated with immediate enlargement of the kidney, with a marked increase in RV pressure, leading to a significant decrease in renal arterial flow. Complications include hemorrhagic infarction, kidney rupture, and retroperitoneal hemorrhage.

Acute RV thrombosis is usually symptomatic with loin, testicular, or flank pain and even the development of scrotal swelling or hydrocele. The patient may present with fever, leukocytosis, and, in the setting of a single kidney or renal transplant, oliguric AKI. Acute RV thrombosis is associated with renal edema and swelling. Nausea and vomiting often accompany it, and symptoms might be confused with those of acute pyelonephritis. Hematuria is nearly universal and most often is microscopic. The high venous pressures result in a marked increase in proteinuria. Urinalysis sometimes reveals evidence of proximal tubule dysfunction, such as glycosuria. In some patients, RV thrombosis is diagnosed only after the patient has developed an acute pulmonary embolus and the source of the embolus is investigated or with worsening of eGFR in the setting of proteinuric CKD.

Chronic RV thrombosis may be asymptomatic. Extensive venous collaterals may allow for minimal impairment of kidney function and structure. Often, however, microhematuria, proteinuria, and evidence of reduced GFR or tubular dysfunction are present, particularly when indices of differential kidney function are sought, such as with nuclear studies. When RV thrombosis causes renal infarction, the distribution of the hypoperfused region tends to be medullary or subcortical. The hypoperfused areas also tend to be patchy and subtotal. These patients can develop severe hypertension acutely.⁹²

The causes of RV thrombosis are listed in [Box 43.8](#).

The prevalence of renal vein thrombosis in nephrotic syndrome is unclear because it is largely undiagnosed; studies report frequencies from 5% to 62%.⁹³ Numerous abnormalities promoting a prothrombotic state occur secondary to heavy proteinuria. It is interesting to note that RV thrombosis appears to be more common in membranous nephropathy and lupus nephritis but can complicate any cause of

BOX 43.8 Causes of Renal Vein (RV) Thrombosis

Malignant neoplasia

- Direct invasion of tumor into the RV
- Retroperitoneal adenopathy, fibrosis, or tumor compressing the RV
- Extension of IVC obstruction by tumor invasion
- Hypercoagulable state associated with malignant disease

Complication of IVC filters

Complication of PICC lines

Nephrotic syndrome

- Membranous nephropathy
- Lupus nephritis

Acute pyelonephritis

Complicating inflammatory bowel disease

Acute pancreatitis

Inflammatory aortic aneurysm

Neonatal

- Congenital
- Dehydration
- Thrombophilia
- Complication of umbilical vein catheterization
- Transmission of maternal procoagulant factors

Hypercoagulable states

- Antiphospholipid antibody syndrome
- Factor V Leiden mutation
- Antithrombin III deficiency
- Protein S and C abnormalities
- Hyperhomocysteinemia
- Elevated levels of clotting factors VIII, IX, and XI
- Heparin-induced thrombocytopenia
- Birth control pill

Thrombophilia

Chuvash polycythemia

Post-renal transplantation

- Acute rejection, OKT3
- Vascular rejection
- Compression or kinking of RV
- Hypercoagulable disorders
- Sticky platelet syndrome
- Calcineurin inhibitors
- Viral infection of the allograft

Complication of surgical compression

- After aortic aneurysm surgery
- After pyeloplasty
- After partial nephrectomy

Traumatic RV thrombosis

Pregnancy

- Compression
- Preeclampsia, eclampsia

Complication of embolization of gastric varices

Budd-Chiari syndrome

Behçet disease

IVC, Inferior vena cava; PICC, peripherally inserted central catheter.

proteinuric kidney disease. In this setting, RV thrombosis can lead to an increase in baseline proteinuria and present with AKI superimposed on CKD.

Pregnancy and the postpartum state are hypercoagulable states. There have been reports of spontaneous RV thrombosis in the postpartum period associated with kidney infarction. Renal vein thrombosis

complicating pregnancy should be suspected when clinical clues such as flank pain, proteinuria, and hematuria are present.

Malignancy accounts for the greatest number of cases of RV thrombosis.⁹⁴ It can result from invasion of a kidney tumor into the RV. About half of renal cell carcinomas are associated with RV thrombosis at autopsy. In addition, neoplasia originating in the RV or inferior vena cava (IVC), such as leiomyosarcoma or cavernous hemangioma, can cause RV thrombosis. Extrinsic compression of the RV by a tumor or retroperitoneal fibrosis also may cause this syndrome.

Diagnosis of RV thrombosis requires imaging. Conventional ultrasound may demonstrate alterations in size and echogenicity. Sonographic findings include kidney enlargement, loss of corticomedullary differentiation, and linear echogenicity radiating from the renal hilum as a result of interlobular and interlobar venous clot. Later scans may show linear, punctuate, or lace-like calcifications in these regions, representing calcified thrombi. Renal duplex ultrasound may show increases in resistive indices and can directly visualize the filling defect. In patients with renal transplant vein thrombosis, the duplex waveform pattern demonstrates reversal of diastolic flow.⁹⁵ Imaging of the RV by magnetic resonance or computed tomographic (CT) venography is needed to confirm thrombosis. In adults, these modalities have much greater sensitivity than RV duplex studies for the diagnosis. Fig. 43.13 is a CT venogram demonstrating unilateral RV thrombosis.

Treatment is controversial and depends on the setting, acuteness, and clinical consequences. If there is no contraindication, most patients are treated acutely with systemic anticoagulation. In adults with acute RV thrombosis that is compromising renal function, catheter-directed thrombolytic therapy with urokinase or tissue plasminogen activator with or without percutaneous mechanical thrombectomy can restore vessel patency.⁹⁶ The long-term benefit of this approach is unclear, and it is less successful when the thrombotic process begins in the small intrarenal venules rather than in the major veins, as is often the case when primary kidney disease or a hypercoagulable state initiates the process.

Surgical interventions include nephrectomy, thrombectomy, and retroperitoneal surgery for non-renal-associated abnormalities, such



Fig. 43.13 Computed tomography venogram demonstrating left renal vein thrombosis (arrow). (Courtesy Dr. S. Rankin, Guy's Hospital, London.)

as tumor, retroperitoneal fibrosis, aortic aneurysm, and acute pancreatitis. Surgery tends to be reserved for situations in which the RV thrombosis results in hemorrhage from renal capsular rupture or for long-term consequences, such as hypertension or infection of a non-functioning kidney.

A conservative approach may be favored when left RV thrombosis occurs because of the extensive collateral venous supply on that side, ultimately allowing venous drainage and delayed improvement in renal function. Systemic anticoagulation is indicated acutely to prevent extension of thrombus into the IVC and pulmonary emboli. Anticoagulation should be continued indefinitely in patients with a persistent hypercoagulable state.

Long-term kidney-related outcomes following RV thrombosis vary with the cause of thrombosis. When associated with membranous nephropathy, it has a more favorable outcome than with other causes. Normal kidney function prior to RV thrombosis is the most important prognostic factor.⁹⁷

SELF-ASSESSMENT QUESTIONS

1. A 33-year-old White woman presents with hypertension that was noted during a regular gynecologic check-up. During examination, you note she has equal BP in both arms, no radial-femoral delay, and a soft abdominal bruit. Which is the most appropriate screening test for the cause of her hypertension?
 - A. RA duplex ultrasound
 - B. CTA of renal arteries
 - C. MRA of renal arteries
2. A 76-year-old woman presents to you because her primary care doctor has had difficulty getting her BP controlled. She states she never had hypertension until about 4 years ago. She has a history of smoking one pack per day of cigarettes for 40 years. She had a femoral-popliteal bypass 2 years ago for claudication. Examination is notable for systolic BP of 180 mm Hg. She has a left carotid bruit audible. Her creatinine is 1.3 mg/dL and potassium is 3.4 mg/dL. A renal duplex ultrasound shows normal velocities in the left RA but a peak systolic velocity of 350 cm/sec in the proximal portion of the right RA. She is on the following antihypertensive medications: metoprolol 50 mg twice daily, amlodipine 5 mg/day, hydrochlorothiazide 25 mg/day, and atorvastatin 80 mg/day, and she is taking aspirin. What is the most appropriate next action?
 - A. Refer for RA stent
 - B. Start ACE inhibitor and follow BP response and potassium and creatinine closely
 - C. Refer to vascular surgery for RA bypass
3. A 32-year-old woman with lupus nephritis class V is currently being treated with induction therapy with mycophenolate mofetil 1500 mg twice daily and oral steroids. She continues to have proteinuria, and her last 24-hour quantitative protein concentration was 6 g. Fortunately, her creatinine has remained normal at 0.8 mg/dL. She presents to the emergency department with pleuritic chest pain. A ventilation/perfusion scan shows high probability for pulmonary embolism. On examination, you note that her creatinine is 1.4 and urine protein/creatinine is 30 g/g. She is started on anticoagulation therapy and admitted. Which of the following tests would you recommend the team order?
 - A. Antiphospholipid antibody
 - B. Renal vein Doppler study and renal ultrasound
 - C. CTA of pulmonary arteries
 - D. Both A and B
4. An 83-year-old woman presents with congestive heart failure and hypertension. She has a history of peripheral vascular disease and chronic kidney disease with a baseline creatinine of 1.5 mg/dL and

a single kidney because she donated one to her son with diabetic nephropathy 30 years earlier. She is admitted and treated with diuretics and achieves a net negative fluid balance of 6 L over 3 days. Her BP, which was 180/90 mm Hg on admission, is now 120/80 mm Hg. An echocardiogram shows left ventricular hypertrophy with diastolic dysfunction, preserved ejection fraction, and no valvular abnormalities. Over the past 24 hours, her creatinine has risen to 3.8, and she has produced almost no urine. She has no flank pain. Despite stopping her diuretics, her creatinine does not improve over the next 72 hours, and she is oliguric. She appears euvoletic and does not respond to empiric intravenous fluids for 24 hours. Her medications are reviewed, and she is receiving no nephrotoxic agents and all diuretics and antihypertensives are on hold. An ultrasound and Doppler examination show a blunted waveform concerning for possible critical RA stenosis. What test would be best to rule out critical RA stenosis?

- A. MRA of RAs
 - B. CTA of RAs
 - C. Captopril renography
 - D. Peripheral renin level
5. A 62-year-old man with a history of hypertension presents with acute left-sided abdominal pain with no dysuria. Urinalysis shows

hematuria without pyuria. He has a temperature of 99.8°F and a white blood cell count of 12,000. He is in new-onset atrial fibrillation. His creatinine is noted to be 1.3, which is above his recent normal level of 0.9 mg/dL 1 month previously. The kidneys appear normal on abdominal ultrasound. Which of the following tests is most likely to provide a diagnosis?

- A. Echocardiogram
 - B. RA duplex ultrasound
 - C. CT with and without contrast
6. A 60-year-old White man with a history of hypertension and diabetes undergoes cardiac angiography with stent placement after presenting to the emergency department with symptoms consistent with unstable angina. Two weeks later, he is noted to have a rise in serum creatinine from a baseline of 0.8 to 1.3. Along with this rise, he is noted to have a slight increase in eosinophils on his complete blood count. On examination the patient is noted to have a serpiginous rash on the bilateral lower extremities to the level of the ankles. Which of the following is the likely cause for the acute kidney injury?
- A. Cholesterol emboli
 - B. Contrast-induced nephropathy
 - C. Acute interstitial nephritis

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